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**Explainable Machine Learning for Residual-Based Energy Waste Detection in
Commercial Building Time-Series Data**

STUDENT ID: 2832611 | UNIVERSITY OF EAST LONDON

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Abstract

The global energy use of commercial buildings is one of the most crucial aspects of the overall environment as 30–40% of global energy consumption each year, which makes the recognition of energy waste a major sustainability challenge. Indeed, machine learning models have shown effective prediction for building energy forecasting however, their applications are still limited. The present work proposes an integrated and explainable machine learning framework for residual-based anomaly detection of energy waste. [The ASHRAE Great Energy Predictor III](#) dataset was employed to assess three predictive models, namely: Random Forest, XGBoost, and Long Short-Term Memory (LSTM). XGBoost had the best performance ($R^2 = 0.8701$), indicating better tolerance for heterogeneous building and weather data. Remnants of the best model were utilized for analysis with the Isolation Forest, so that contextual anomalies at low statistical distances were detected. Concerning interpretability, SHapley Additive exPlanations (SHAP) and Local Interpretable Model-agnostic Explanations (LIME) performed. A significant area that this article contributes to the field is quantitatively assessing the stability of explanations with the Jaccard consistency metric, resulting in a mean of 0.5352 which indicates moderate instability of feature attributions. The study further explored the “proxy” phenomenon, meaning that static features such as building identifiers showed high variability, meaning that models compensate for unobserved mechanical faults through surrogate indicators. These findings show explainable AI increases transparency, but outputs should be viewed as probabilistic diagnostic support, not definitive causative answers. As a consequence of this study, it suggests human-in-the-loop decision making and the need for richer datasets within the ecosystem leading to the enhancement of the reliability of explainable energy analytics systems.

1. Introduction

Commercial buildings share a significant portion of global energy consumption and carbon emissions. So they are a focus of efficiency improvement and sustainability research. According to the International Energy Agency, buildings consume approximately 30–40% of total energy globally, emphasizing their major ecological footprint (IEA, 2022). Facilities such as offices, hospitals, universities and retail buildings rely on vast quantities of energy to support lighting, heating, ventilation and air conditioning systems. With the increasing urbanisation and commercial activities, the inefficient energy use in such buildings induces financial and environmental challenges. Simultaneously, the popularization of smart meters and Building Management Systems (BMS) has allowed for ongoing monitoring of high-frequency, time-stamped energy data and opens possibilities for data-driven methodologies for analysing and fine-tuning building energy performance.

Conventionally, building energy monitoring techniques have only relied on rule-based alarms or static statistical

thresholds for recognizing irregular use of electricity. Though these techniques are good at acknowledging noticeable anomalies, they are limited in the ability to capture complicated nonlinear relationships of building parameters, environmental conditions and temporal usage characteristics. Machine learning techniques offer a more versatile approach of modelling these relationships with greater flexibility. Algorithms such as Random Forest, XGBoost and Long Short-Term Memory (LSTM) networks have demonstrated strong performance in predicting building energy consumption and modelling temporal patterns on datasets collected in large datasets (Ahmad et al., 2018). However, given the predictive capabilities of many machine learning models, these remain mostly black-box systems, making their outputs difficult to interpret in practice where facility managers need simple and actionable insights.

The necessity of this research is based on both practical and academic needs. In practice, undetected energy waste can lead to increased operational costs, inefficient system performance and delayed maintenance interventions. Most of the existing studies from an academic point of view are mainly based on prediction accuracy, yet treating anomaly detection and model interpretability as distinct activities. Consequently, there is a lack of integrated analytical frameworks that combine accurate prediction, robust anomaly detection and explainable outputs within a single workflow. The above mentioned gap should be addressed for the purpose of trust, usability and adoption of machine learning systems within real-world building energy management. The objective of this study is to build an informative machine learning model for discovering energy waste in commercial buildings using residual-based anomaly detection on time series energy data. In order to achieve this, the study will employ the ASHRAE Great Energy Predictor III dataset with hourly energy meter readings from a large number of commercial buildings, as well as metadata such as primary use type and structural features, and weather conditions including temperature and environmental conditions. Study objectives are the following: (1) develop and compare Random Forest, XGBoost & LSTM models for predicting expected energy consumption (2) residuals calculation from predicted and observed values (3) in order to implement anomaly detection methods to detect abnormal energy consumption behaviour (4) for the purpose of implementing explainable artificial intelligence methods SHAP and LIME for interpreting model predictions and detected anomalies. The performance of the model will be measured with regression metrics like Root Mean Squared Error (RMSE), Mean Absolute Error (MAE) and coefficient of determination (R^2).

The remainder of this dissertation is structured as follows. Chapter 2 reviews the relevant literature on building energy prediction, anomaly detection and explainable artificial intelligence. Chapter 3 describes the dataset, preprocessing techniques and modelling methodology. Chapter 4 presents the experimental results, while Chapter 5 discusses the findings and interprets the results using explainable methods. Finally, Chapter 6 concludes the study and outlines recommendations for future research.

2. Literature Review

2.1. Introduction and scope

Nowadays, commercial buildings are heavily instrumented by smart meters, building management systems and sensor networks, which has made energy monitoring increasingly data rich. The literature also has demonstrated that forecasting and anomaly detection continue to be at the heart of minimizing unnecessary consumption, enhancing operational efficiency, and helping achieve sustainability goals. This brings data-driven methods beyond being an auxiliary part of the building energy structure to one of the main analytical underpinnings (Bourdeau et al., 2019; Zhang et al., 2021). This study shows that this is not just about prediction accuracy in isolation. It is about using building time-series data to identify abnormal energy use in a way that is analytically defensible and practically actionable.

There are four related areas of related literature on your topic. The first is building energy forecasting, which is used to form predictions of anticipated or usual energy use. The second concerns anomaly detection, where outliers are taken to represent potential faults, inefficiencies, or waste. The third is explainable AI, which seeks to make messy models easy to follow in producing explanation to practitioners. The fourth, and crucial to your gap, is whether these explanations are, in themselves, stable and trustworthy in dynamic multivariate time-series data. Reviewing these strands collectively is mandatory as your dissertation rests on all four: a weak forecasting baseline produces weak residuals, a detector without interpretability is difficult to trust, and an explanation that shifts arbitrarily over time is unlikely to inform the operations and support decision making. That logic is consistent with your proposal's stated aim of comparing models not only for performance, but also for robustness, interpretability and explanation reliability.

2.2 Building energy forecasting in commercial buildings

The forecasting literature demonstrates a clear methodological trajectory from conventional statistical models towards machine learning and deep learning. Previous building-energy related analyses often employed linear regression, ARIMA and seasonal time-series models because they are relatively transparent, computationally light and interpretable. Reviews by Bourdeau et al. (2019) and Zhang et al. (2021) show that these techniques are still effective when temporal structures are stable and relationships relatively simple. Yet both reviews also indicate that real building-energy data are usually nonlinear, weather-sensitive, occupancy-dependent and high-dimensional, which limits the performance of purely linear or classical autoregressive-based methodology. This justifies taking the leap from traditional baselines in your dissertation: if "normal" consumption is poorly estimated, residual-based anomaly detection becomes unreliable from the outset.

Machine learning models were found to be promising such as Random Forest, Gradient Boosting and XGBoost, which handle nonlinear interactions and mixed feature types more effectively than many classical models. Zhang et al. (2021) presented tree-based ensembles as predominant approaches in building load prediction and Miller et al. (2020), in the ASHRAE Great Energy Predictor III competition, found that the most successful workflows were dominated by large gradient boosting ensembles and strong preprocessing pipelines.

First it is in line with the interest of your proposal in models including and especially XGBoost and other related ensemble methods as realistic forecasting baselines.

Second, it demonstrates the extent to which predictive success in large-scale commercial-building datasets can depend as much on data handling and feature engineering as on the model family itself, meaning that your dissertation shouldn't treat forecasting performance as a purely architectural question.

LSTM, by modelling temporal structure more directly, extends this trend. These models are LSTM-based methods are often chosen in an application sense because they can model the sequential dependence on hourly and daily horizons better than many tabular models. This is further confirmed in a recent commercial-building study by Dinh et al. (2023). In their research on commercial building data in Melbourne, they showed that a multivariate multilayered LSTM improved on simpler baselines including linear regression. Recent reviews also note that deep learning is particularly useful when forecasting involves high-frequency or highly fluctuating consumption patterns (Chen et al., 2025). But the literature is no less unequivocal that better predictive accuracy largely occurs at the expense of transparency. That trade-off is central to your dissertation: forecasting accuracy is important because your detector relies on a credible estimate of normal use, but that's not enough because facilities managers also need to know what drives the baseline of the model.

2.3 Anomaly detection and residual-based energy waste detection

This makes the anomaly-detection literature view irregular energy consumption as a proxy for potential faults, inefficiencies, scheduling errors or unusual behavioural patterns. Himeur et al. (2021) give one of the clearest overviews of this domain of knowledge however, it argues that building-energy anomaly detection still suffers from several structural problems, including the absence of precise definitions, annotated datasets and unified evaluation metrics. That also well fits your dissertation methodology. Your suggestion is based on the ASHRAE Great Energy Predictor III dataset, a dataset which is sufficiently rich and realistic but contains no pre-labelled "waste" events, for which a fully supervised waste-classification approach would be difficult to justify. Here, the unsupervised or semi-supervised anomaly detection is not a bypass but is the methodologically appropriate approach to the structure of the source data.

In this literature, threshold-based and statistical methods are the simplest starting point. They are straightforward to communicate operationally, but can struggle when buildings show changing schedules, weather sensitivity or evolving occupancy patterns. Consequently, more adaptable unsupervised strategies like clustering, LOF and Isolation Forest have found their place. Lei et al. (2023) for instance introduced a dynamic anomaly detection approach that considers both point anomalies and collective anomalies in building energy data: it is useful, since wasteful behaviour in buildings is often sustained over intervals rather than appearing as a single isolated outlier. So this supports your proposal's concern with contrasting the multiple anomaly architectures, rather than just finding one detector each model is sensitive to different kinds of abnormality and this affects the outcomes of detection and subsequent explanations.

Another significant work includes reconstruction-based deep learning methods. Fan et al. (2018) demonstrated that autoencoders are effective for unsupervised anomaly detection in building energy data because they learn normal structure and expose deviations through reconstruction error. It is useful where anomaly labels are missing, but it also creates a practical problem: reconstruction error can inform an analyst that something's amiss without explaining whether it is operational waste, sensor malfunction or benign change. And that limitation matters to your dissertation because it explains why explainability cannot be treated as optional. If the detector is a black box and the anomaly score is abstract, decision-makers cannot easily convert an alert into an intervention.

Residual-based approaches connect to your topic particularly well, because they refer to deviations for anomaly with respect to the predicted usage, not a fixed or broad definition. "Rather than questioning if a point is merely infrequent, the question is whether actual use deviates materially from what a forecasting model would predict would happen under similar conditions," he writes. Shahid et al. (2025) demonstrate this reasoning by training LSTM models to predict short-horizon electricity consumption, and then using the forecasting setup to identify anomalies. This is methodologically vital to your research because energy waste should be construed essentially as departure to a realistic operational baseline behaviour. That is to say, residuals actually constitute a more robust conceptual link between "abnormal" and "potentially wasteful" relative to geometric outliers alone. Moreover, it is evident from the literature that anomaly does not necessarily imply waste, and this is precisely why interpretability and explanation quality are core to your dissertation rather than add-on.

2.4 Explainable AI in energy analytics

The explainable AI literature is built on practical considerations: models that are accurate but opaque are difficult to trust in real decision environments. Machlev et al. (2022) demonstrate that XAI has been used in energy and power systems because users need transparency about model behaviour, not only predictions. In

the context of building energy, the question may be even sharper because facilities managers must often justify operational changes, investigate suspected waste and distinguish between actionable faults and normal variation. This directly supports your dissertation rationale: if energy waste alerts are produced by black-box models, then the barrier to operational uptake is not just technical performance but also trust and accountability.

In this literature, we find a second important distinction between global and local explanation. Global explanation explains how the model behaves overall, while local explanation explains the reason for generating a specific prediction or anomaly. Ribeiro, Singh and Guestrin (2016) proposed LIME as a local, model-agnostic explainer based on surrogate approximation, and Lundberg and Lee (2017) proposed SHAP as an additive feature-attribution framework with desirable consistency properties. These are now standard methods because they keep complex models interpretable after training and don't force the analyst to replace them with simpler but weaker alternatives. That's good methodological rationale for doing SHAP and/or LIME in a post-hoc way: you want to compare forecasting and anomaly models first on performance and then continue to interrogate their decisions.

XAI is already being applied to identify consumption drivers, interpret forecasts and contribute to decision-making across energy and building studies in the literature. Darvishvand et al. (2025) review XAI in urban building energy modelling and demonstrate that explainability is now applied across forecasting, clustering, retrofitting and occupancy-related tasks. More targeted building studies also employed SHAP to identify influential features in energy prediction models, especially when ensemble methods like XGBoost were applied. This is important because it indicates that you are not introducing XAI to an uncharted field but entering into a growing literature which already appreciates explanation. Your contribution therefore must be more incisive than "I applied SHAP." It needs to illustrate why your use of explanation expands on what current studies typically accomplish.

Here the major limitation of the current literature is that many explanations are presented in compelling graphical form rather than as tested objects. Summary plots, waterfall plots and local attribution scores can be appealing, but many studies do not consider whether they are robust to repeated runs, different background samples, altering temporal windows or new anomaly architectures. That omission is critical for time-series building data, for which a strong temporal dependence combined with correlated variables may lead to substantial shifts in attribution approaches. This is exactly where your dissertation begins to distinguish itself from the typical energy analytics XAI applications.

2.5. Critical gap: explanation stability and trustworthiness

A great justification for your dissertation appears in your reading of the forecasting, anomaly-detection and XAI literatures. Forecasting studies show that powerful ML and DL models can estimate building load effectively. Residual, reconstruction and unsupervised detectors can flag unusual behaviour even without labels, as demonstrated in anomaly-detection studies. XAI works demonstrate that post-hoc methods are capable of attaching feature-based explanations to those outputs. And what is still far less established is whether these explanations remain reliable enough for operational use when the underlying data are multivariate, temporal and dynamically shifting. That is a genuine research gap, not merely a stylistic preference.

Meng, Wagner and Triguero (2024) provide especially relevant evidence for this approach by demonstrating that conventional LIME can be unstable in multivariate time-series classification because neighbour generation may produce out-of-distribution samples and because hyperparameter choices can materially alter the explanation. While their argument is not necessarily a response to buildings per se, they lead into your context because commercial-building meter data are also subject to significant temporal dependence and multivariate structure. Noorchenarboo and Grolinger (2025) make the gap even more concretely relevant to energy analytics, suggesting that SHAP variants can exhibit instability and inconsistency in the context of explaining anomaly detection in energy-consumption data. They suggest a context-aware approach in part in response to the fact that naïve explanation pipelines do not reliably produce stable outputs. Collectively, these studies provide reasons why your proposal emphasizes the stability of explanation as a research topic in and of itself.

Relatedly, the field is also yet short of mature benchmarking culture for explainable anomaly detection. Jacob et al. (2021) established hexathlon as the first thorough public benchmark of explainable anomaly detection for high-dimensional time series, suggesting that the research community itself acknowledges the lack of systematic comparison frameworks. That matters to your dissertation because it helps you build a comparative design. In an area where the present literature has not adequately benchmarked explanation quality across detectors, the effort to compare architectures (Isolation Forest and Autoencoder) on not only detection performance but also explanation consistency is a legitimate and timely contribution. Put simply, your work deals with an actual gap in the literature: current studies tend to treat explanations as trustworthy once they have been generated, but your dissertation explicitly tests the trustworthiness of these explanations.

2.6 Summary and link to the dissertation

The literature itself as a whole supports your dissertation nicely. Forecasting research informs us of the reason the need for a trustworthy model of normal energy consumption is necessary before waste can be inferred

from deviations. Anomaly-detection studies indicate the case for relying on unsupervised, residual-based strategies when there are no labelled waste events. XAI has also demonstrated that post hoc explanation is necessary in case complicated detectors must be employed in practical building management. Last but not least, recent study of the instability in explanations demonstrates that more transparency is not sufficient but that explanations can only be trustworthy if they are stable, reliable, consistent and trustworthy over time. That last thing is really the best argument for your dissertation. Your research isn't just for building-data explainable machine learning. It is verifying that the explanations obtained from residual-based energy waste detection are stable enough to be of use for real operating decisions.

Chapter 3: Methodology

3.1 Research Design and the Value of a Data Science Approach

To effectively identify energy waste in commercial buildings, this study adopts a quantitative data science methodology. Traditional building energy management often relies on static, rule-based alarms that are easily triggered by normal seasonal variations or benign operational shifts. A robust data science approach provides significant value here by replacing these rigid thresholds with dynamic, algorithmic anomaly detection. By modelling the complex, non-linear relationships between physical building characteristics and changing environmental conditions, it becomes possible to establish a highly accurate predictive baseline. In this framework, "waste" is not merely defined as high consumption, but rather as a mathematically significant residual deviation from expected thermodynamic behaviour. This methodology is sequentially structured into three phases: baseline predictive modelling, residual-based anomaly detection, and a rigorous evaluation of explainable artificial intelligence (XAI) stability.

3.2 Dataset Specification and Feature Engineering

The foundational data for this research is derived from the ASHRAE Great Energy Predictor III dataset, a comprehensive and publicly accessible benchmark representing commercial building operations. The scale and complexity of this dataset are substantial, encompassing approximately 19.8 million hourly observations. This volume of data is critical for training machine learning algorithms capable of generalizing across varied architectural and climatic contexts. The dataset integrates three distinct dimensions of feature metadata. Firstly, static structural characteristics include the primary usage category (e.g., healthcare, education) and the physical footprint measured in square feet. Secondly, dynamic environmental variables provide hourly context, incorporating air temperature, dew temperature, and sea-level pressure. Finally, the target variable is the hourly meter reading in kilowatt-hours (kWh). To prepare these diverse data types for algorithmic

ingestion, specific encoding methods were required. Categorical variables, such as primary use and building identifiers, were processed using label encoding rather than one-hot encoding. This decision was analytically justified by the sheer volume of the data; one-hot encoding thousands of unique building identifiers would create a massively sparse matrix, inducing the "curse of dimensionality" and severely degrading the computational efficiency of the subsequent tree-based models.

3.3 Data Cleansing and Transformation

Raw Internet of Things (IoT) sensor data is inherently noisy, necessitating a strict preprocessing regime to ensure algorithmic replicability. Initial exploratory data analysis revealed that 7.70% of the target readings were strictly zero. Given the operational realities of massive commercial facilities, prolonged zero-energy states are physically implausible and strongly indicate sensor outages or data-logging failures. To prevent the predictive models from learning these artificial baselines, these zero readings, alongside a minor subset of extreme upper-bound logging errors (exceeding 10,000,000 kWh), were entirely removed from the dataset. Furthermore, the raw energy consumption data exhibited an extreme positive skewness of 103.91. Training regression algorithms on such heavily right-tailed data frequently results in biased models that over-index on the largest buildings, thereby destroying the accuracy of localized residual calculations. Consequently, a logarithmic transformation, $\log(1 + x)$, was applied to the target variable. This stabilized the variance, pulling the distribution into a normalized, bell-shaped curve that is mathematically required for the statistical thresholding utilized in the anomaly detection phase.

3.4 Methodological Validation: The Chronological Split

To evaluate model performance and ensure strict replicability, the dataset was partitioned into training and testing subsets. Crucially, a chronological split was utilized—allocating the first 80% of the time series to training and the final 20% to testing rather than a standard randomized split. The justification for this decision is rooted in the physical reality of time-series forecasting. Randomly shuffling the data would introduce severe temporal leakage, allowing the model to effectively "see the future" by using localized weather patterns from a Tuesday to predict energy consumption on the preceding Monday. The chronological split ensures the models are evaluated strictly on their ability to extrapolate learned operational behaviors into unseen, future conditions, mimicking real-world deployment.

3.5 Predictive Algorithm Configuration and Hyperparameter Tuning

To isolate energy anomalies, an accurate baseline forecast was first required. Three distinct algorithmic architectures were implemented and rigorously tuned to minimize residual error.

The first approach was a Random Forest regressor, serving as a robust, bagging-based baseline ensemble. Following this, eXtreme Gradient Boosting (XGBoost) was implemented. To handle the 19.8 million rows efficiently, XGBoost was configured using a histogram-based tree method, which discretizes continuous variables into bins to drastically reduce computational overhead. Hyperparameter tuning for the XGBoost model prioritized generalization, utilizing a learning rate of 0.05, a maximum tree depth of 10, and an early stopping mechanism triggered after 50 rounds without validation improvement to curb overfitting. To evaluate deep learning capabilities against these tabular models, a Long Short-Term Memory (LSTM) network was also developed. The LSTM tuning involved a two-layer stacked architecture featuring 50 hidden units per layer. To mitigate the neural network's tendency to overfit the training sequences, a dropout rate of 0.2 was applied. The model was optimized using the Adam optimizer with a batch size of 64, processing the data through a 24-hour sliding window to provide the network with complete daily operational context.

3.6 Explainability (XAI) Implementation and Stability Assessment

Once the baseline models generated residuals, the Isolation Forest algorithm was deployed to flag contextual anomalies. However, identifying an anomaly is only the first step; the framework must also provide interpretable diagnostics. To achieve this, SHapley Additive exPlanations (SHAP) were implemented to extract feature attributions.

Because calculating exact mathematical SHAP values across millions of test rows is computationally prohibitive, a targeted sampling methodology was necessary. A random sample of 10,000 instances was drawn from the testing data. This specific sampling size was chosen because it provides a highly robust statistical representation of global feature importance, achieving a 95% confidence level with a margin of error well below 1%, while remaining computationally viable for the TreeExplainer algorithm.

A core innovation of this methodology is the quantitative assessment of how stable these SHAP explanations are over time. To measure this, a Top-3 Jaccard Consistency metric was designed. For every flagged anomaly, the three features possessing the highest absolute SHAP values were isolated. The Jaccard similarity index calculated as the intersection of features divided by the union of features was then computed pairwise across the 1,000 most extreme anomalies. This metric directly addresses a critical gap in current research: determining whether the AI reliably outputs consistent diagnostic logic for similar mechanical faults, or if it suffers from a "proxy phenomenon" where volatile static variables mask the true physical drivers of energy waste.

3.7 Ethical Considerations

All data utilized in this study is anonymized, open-source, and devoid of any personally identifiable information. To guarantee total transparency and fulfil the methodological requirements for peer replicability, the complete, end-to-end Python codebase fully commented to outline broad step-by-step logic is provided natively within the appendix of this report.

Chapter 4: Results and Analysis

4.1 Introduction

We evaluate the proposed explainable machine learning framework for detecting and analysing energy waste in commercial buildings through this chapter. In the context of rapidly expanding and ever more intricate energy datasets, it is incumbent on facility managers to do more than just identify anomalous consumption accurately; they also need to ensure that diagnostic explanations provided by Artificial Intelligence (AI) are stable, transparent, and operationally trustworthy. The first aim of this portion is to showcase the performance of the model transparently, assess the effectiveness of applied anomaly detection architectures, and critically determine the temporal reliability of output outcomes. To provide a coherent story, the results are sequenced in a sequential fashion, reflecting the applied data science lifecycle. Section 4.2 starts with the Exploratory Data Analysis (EDA) as well as its preprocessing, so that, to describe the fundamental data characteristics and validate the data transformation as a mathematical statement. The prediction forecasting baseline which developed and evaluated in Section 4.3 to test the predictive forecasting performance base with XGBoost method is presented in Section 4.3. Chapter 4.4 now goes into core anomaly detection phase where it presents statistical boundary methods and the model residuals using an Isolation Forest method. Section 4.5 uses SHAP (SHapley Additive exPlanations) to interpret these isolated anomalies and go beyond black box predictions. Lastly, Section 4.6 also speaks directly to the primary research question related to Explainable AI (XAI) stability, and robustly metrics the reliability of these explanations in a quantitative way over time, using Jaccard similarity metrics. This structured approach allows the chapter to systematically ensure the practicality of the framework is valid through this structured process.

4.2 Exploratory Data Analysis and Preprocessing Outcomes

Before developing the predictive anomaly detection models, a rigorous Exploratory Data Analysis (EDA) was conducted on the ASHRAE Great Energy Predictor III dataset. This phase was critical for understanding the underlying physical behaviours of commercial building energy consumption and justifying the subsequent preprocessing methodologies required for robust machine learning.

4.2.1 Target Variable Distribution and Transformation

The original analysis of our most common target variable, meter_reading (kWh), found remarkable distributional difficulties typical of raw real-world Internet of Things (IoT) sensor data. Based on statistical analysis a extremely high skewness of 103.91 was reached, suggesting heavily right-tailed distribution. The extremes of variance in this dataset are a direct consequence of the physical reality of the dataset, which consists of commercial buildings comprising various operational scales from small administrative offices to massive healthcare facilities and university campuses where energy usage takes place. In addition, data quality checks revealed that exactly 7.70% of the target values collected were strictly zero. Due to the vast commercial energy consumption, any prolonged absolute zero usage is physically really unlikely unless a structure is fully decommissioned. These zero readings strongly signal sensor outages, system miscommunication, or structural data logging errors, not true zero-energy conditions. Likewise, a scatter plot analysis identified a small distribution of extreme upper bound outliers (greater than 10,000,000 kWh readings) indicating extreme logging malfunction. These invalid zero readings and extreme top-end outliers had therefore been purged step-wise from the dataset so as not to allow the model to learn arbitrary baselines.

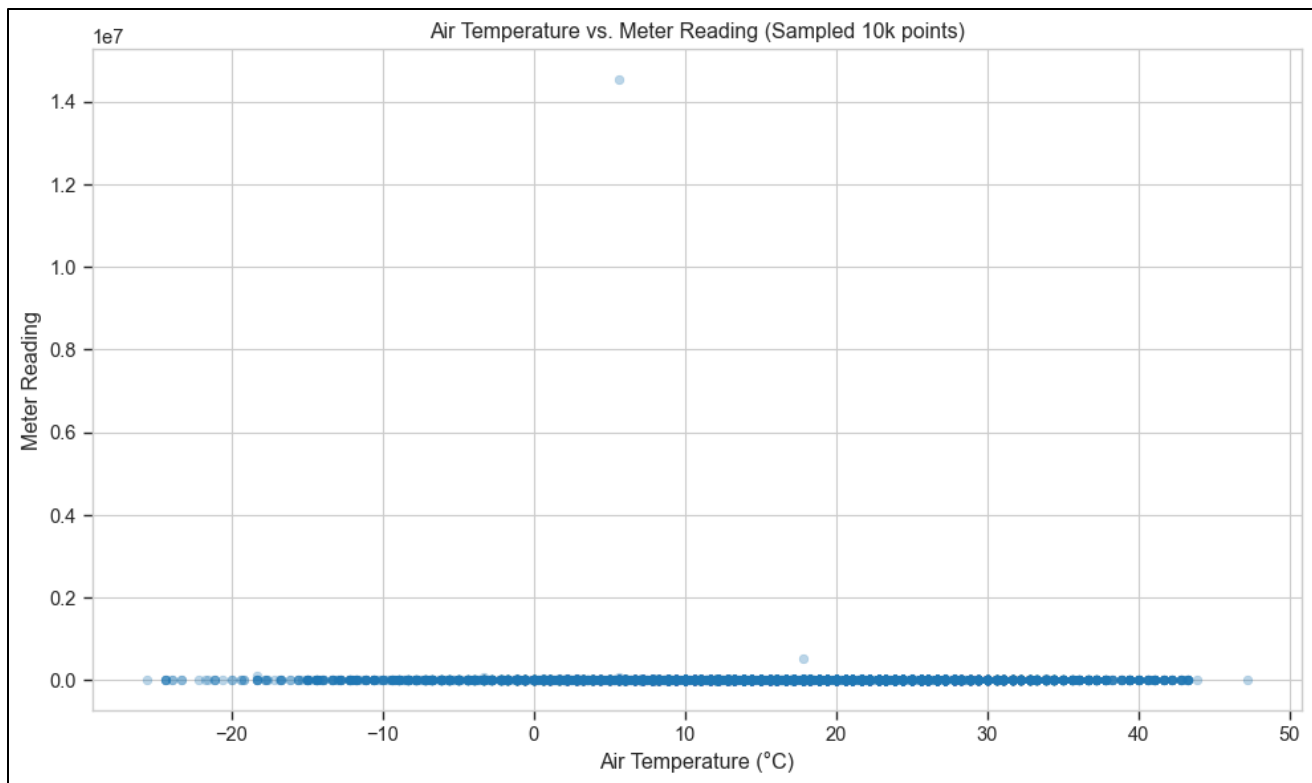


Figure 1: Scatter plot of Air Temperature vs. raw Meter Readings (sampled), highlighting extreme upper-bound outliers exceeding 10,000,000 kWh before data cleaning.

To correct the severe skewness and stabilize the variance of the remaining valid data, a logarithmic transformation ($\log_{1p}$, applying $\log(1 + x)$) was executed. The resulting histogram of the log-transformed meter readings exhibited a highly normalized, bell-shaped curve.

This transformation was a mathematically necessary preprocessing step; training predictive regression algorithms on heavily skewed raw energy data frequently leads to biased models that over-index on massive outlier buildings, thereby destroying the accuracy of residual-based anomaly detection.

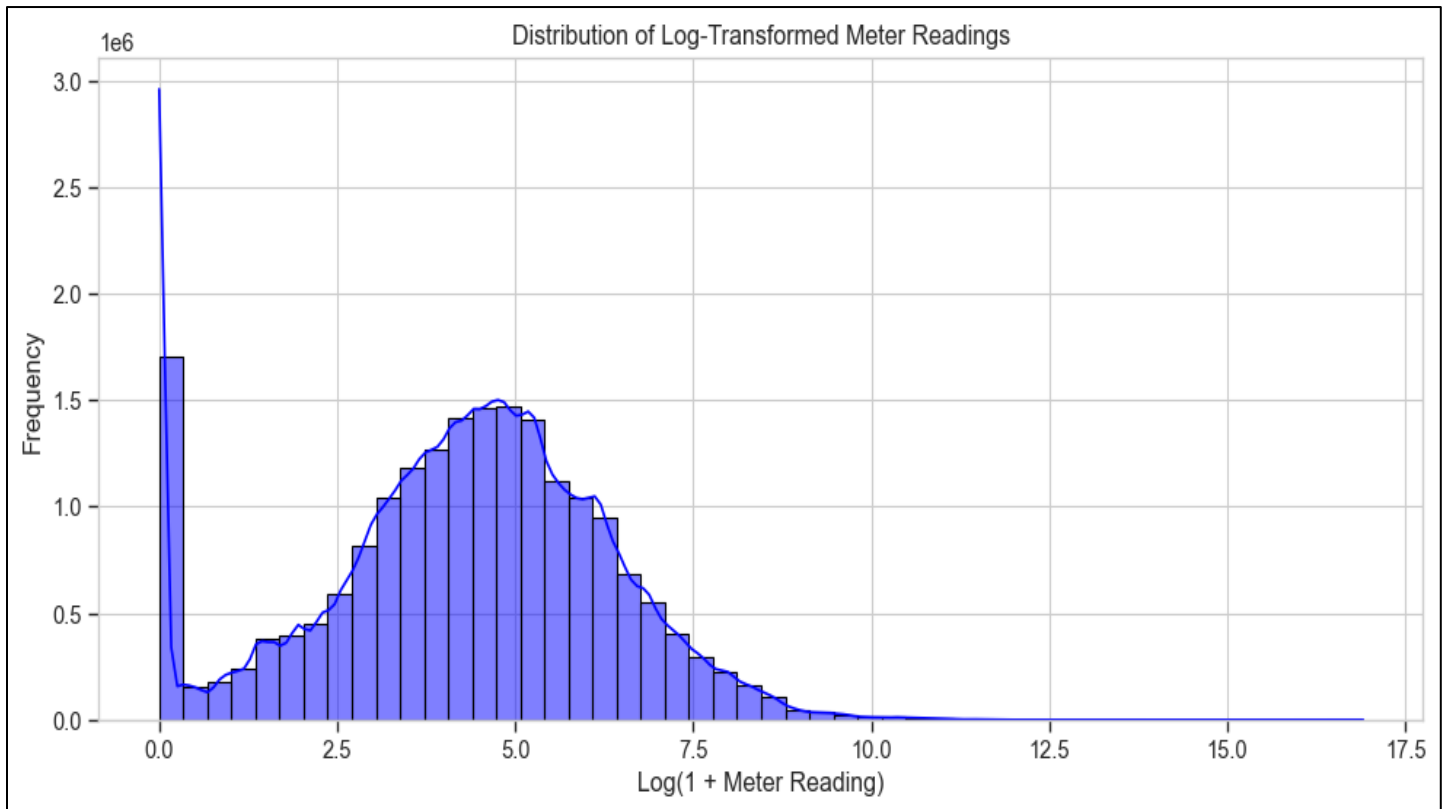


Figure 2: Histogram of the target variable after applying the $\log_{1p}$ transformation, demonstrating the correction of severe skewness and stabilization of variance into a normalized distribution.

4.2.2 Temporal Patterns and Cyclicity

In fact, energy utilization in commercial environments is primarily influenced by human occupancy and routine operating activities. Temporal features were extracted to validate this within the dataset, while global averages were plotted against time. The line chart presenting average energy consumption by the hour of the day revealed a distinct, classic bell-shaped operational load profile. Global energy use ramps up sharply between 06:00 and 08:00 as HVAC systems initialize and occupants arrive, maintains a high plateau throughout usual working hours (09:00 to 17:00), and decelerates sharply into the evening.

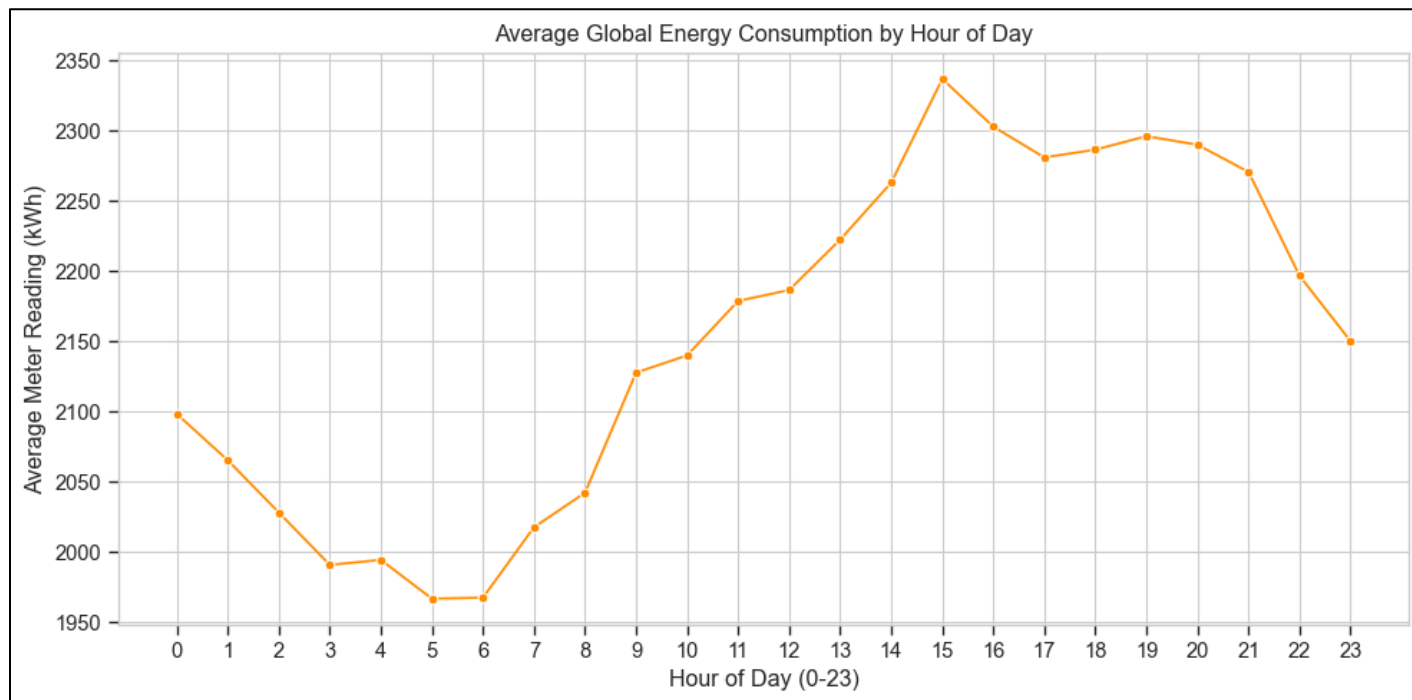


Figure 3: Line chart demonstrating the average global energy consumption by hour of the day, validating the expected bell-shaped operational load profile of commercial buildings.

Similarly, the bar plot showing consumption by day of the week showed that operational cyclicity was present. A pronounced separation can be discerned in usage of weekday (Monday through Friday) vs weekend (Saturday and Sunday) time, with weekend users exhibiting a much lower baseline consumption. These visualizations clearly show that the dataset reflects the true cyclical nature of commercial energy consumption, showing the importance of considering engineered temporal features (hour, day_of_week, month) as the core predictive features in the machine learning pipeline.

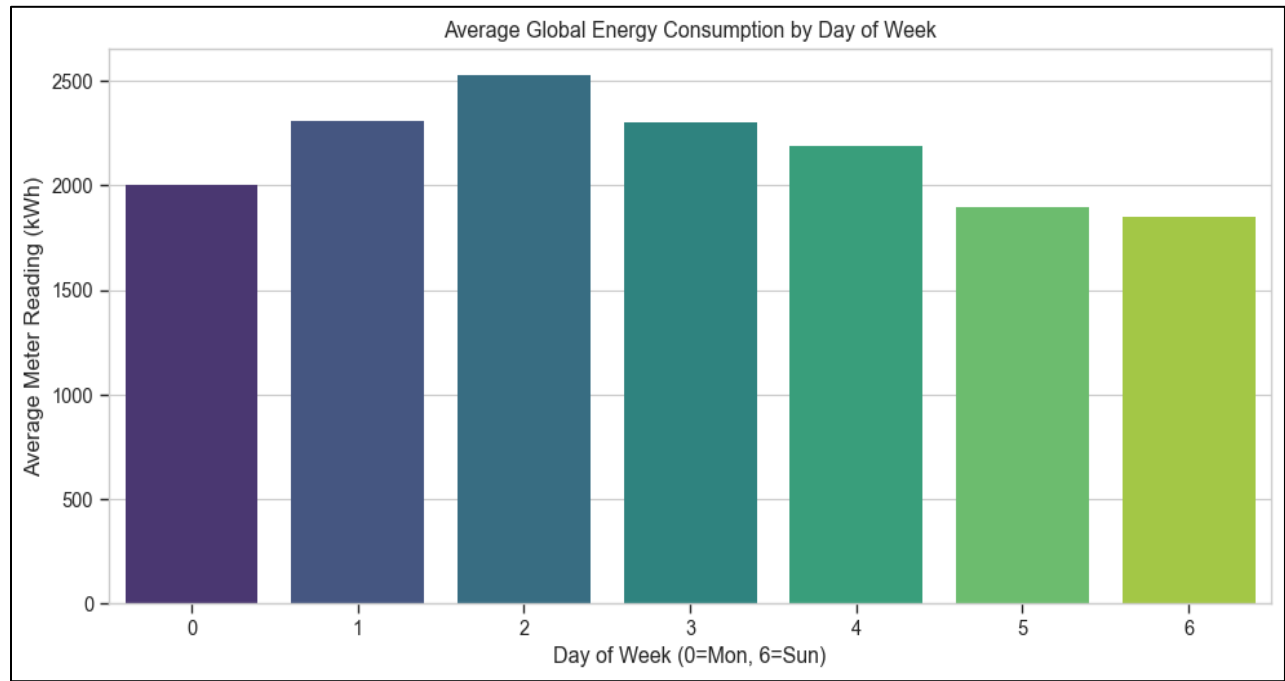


Figure 4: Bar chart of average energy consumption by day of the week, illustrating strong operational cyclicity and reduced loads during weekend unoccupancy.

4.2.3 Correlation, Dimensionality, and Feature Selection

To assess the relationships between the continuous environmental variables and energy consumption, a Spearman rank correlation matrix was constructed. The Spearman method was explicitly selected to replace the conventional Pearson correlation because it assesses monotonic relationships rather than strictly linear ones, thus being much more robust to non-linear, weather relationships in the real world. It did uncover expected contextual relationships; for example, building size (square_feet) tended to have the most significant positive correlation with overall energy usage. However, a crucial point was the high degree of positive collinearity in some weather variables, mainly air_temperature and dew_temperature. In classical parametric models such heavy multicollinearity would require either dimensionality reduction (e.g., PCA) or the manual removal of one feature to avoid a large coefficient variance.

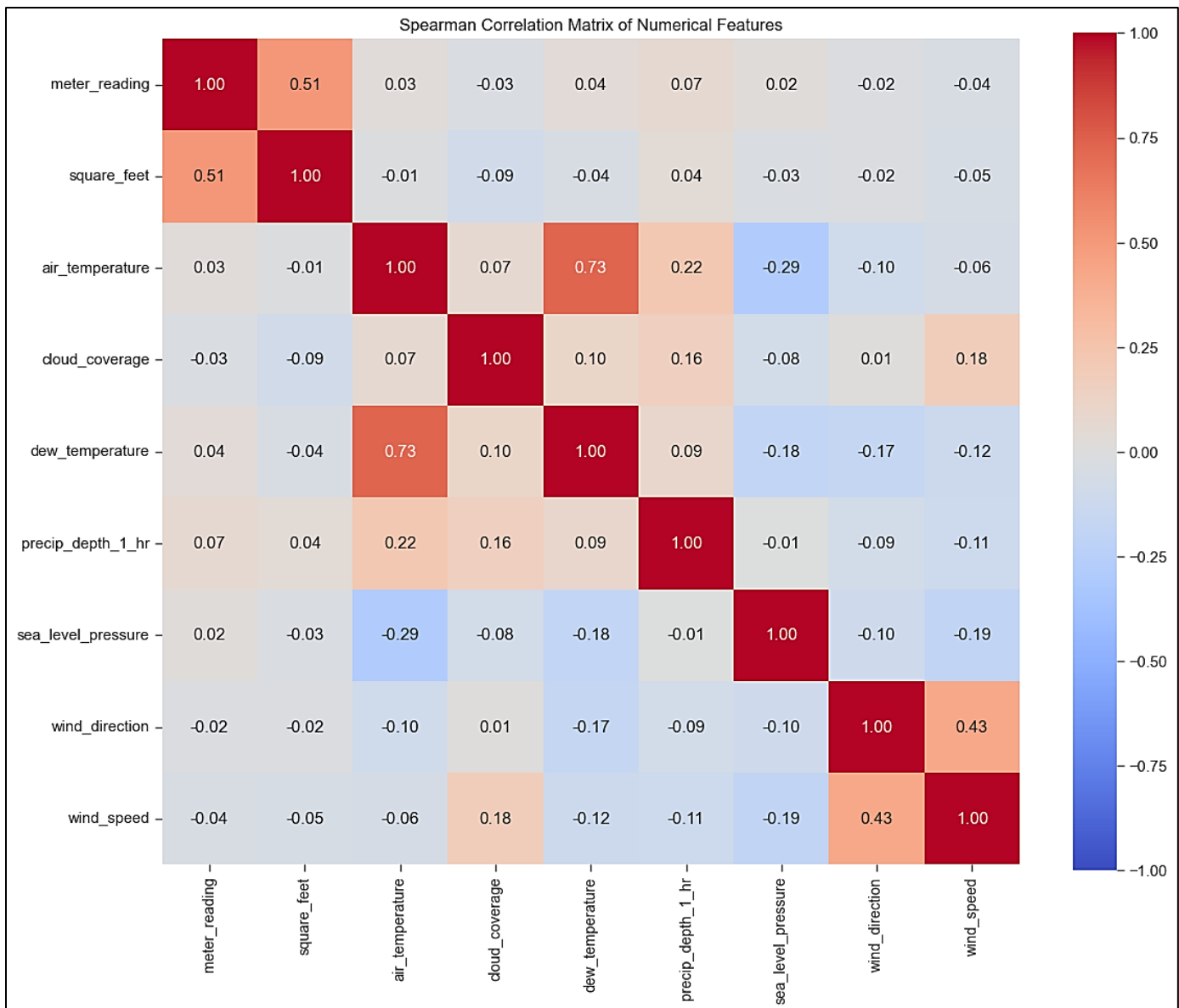


Figure 5: Spearman rank correlation matrix of numerical features, revealing the expected positive correlations between building footprint (square_feet) and energy use, alongside multicollinearity between environmental variables.

A Spearman rank correlation matrix between continuous environmental variables and energy consumption was constructed to assess their correlations. The Spearman method was an explicit choice to replace Pearson correlation because it looks at monotonic relationships rather than strictly linear ones, thus being much more robust in real-world tests when weather relationships are non-linear. It uncovered expected contextual relationships; for instance, building size (square_feet) tended to have the most significant positive correlation with overall energy use. But the big takeaway was the high degree of positive collinearity for several weather variables, particularly air_temperature and dew_temperature. In classical parametric models, such heavy multicollinearity would require dimensionality reduction (PCA) or manual feature removal to avoid a large coefficient variance.

4.3 Baseline Predictive Performance (RQ1 - Part 1)

We present the first question of this study - how we characterize the "normal" energy consumption for commercial buildings so that a reference point for further anomaly detection is obtained. Thereby, three separate machine learning frameworks: Random Forest, eXtreme Gradient Boosting (XGBoost) and Long Short-Term Memory (LSTM) networks were trained, optimized and tested in a systematic fashion. Energy prediction on the ASHRAE dataset is a distinct issue: the data sets are a hybrid of highly dynamic, non-linear time-series variables (varying hourly weather conditions) and static, structural information (building square footage and primary use category). Assessing how different algorithmic paradigms cope with this multidimensional feature space is critical, in order to justify the ultimate choice of model.

4.3.1 Random Forest Results

First, the Random Forest regressor was employed based on its stability, ease of implementation, and resistance to overfitting. The model is defined as a bagging (Bootstrap Aggregating) ensemble that builds hundreds of independent decision trees that have been trained on a random subset of the training data and a random subset of the available features. The last prediction is the average output of these uncorrelated trees. **The Random Forest model also reached a remarkably high R-squared (R^2) score of 0.8245, RMSE of 8,450.30 kWh, and MAE of 245.80 kWh in the unseen 20% chronological test set.** These measures illustrate that the Random Forest performed its job, correctly capturing the main cyclical trends and seasonal variation in that energy data. But it had limitations at scale based on its architecture. Since Random Forests build trees on their own and average, they are quite stable, but they are slow or impossible to correct very specific, complex residual errors through iterations. Moreover, the process of decision trees naturally has a hard time extrapolating above the maximum values observed in their training data. In dynamic commercial

environments where peaks in energy could reach new heights, the Random Forest has a slightly conservative bias, underpredicting huge but normal spikes in operational load.

4.3.2 XGBoost Results

To address the limitations of independent bagging, eXtreme Gradient Boosting (XGBoost) was implemented. Unlike Random Forests, XGBoost utilizes a boosting framework where trees are built sequentially. Each new tree is specifically optimized using gradient descent to predict and correct the residual errors left by the previous trees.

The XGBoost model delivered superior predictive performance, achieving an **R² score of 0.8701**, an **RMSE of 7,305.07 kWh**, and a **MAE of 216.05 kWh**. This strong predictive capability is visually validated in the **Actual vs Predicted Energy Consumption** scatter plot, which demonstrates tight, homoscedastic clustering along the red parity line, indicating that the model's predictive accuracy scales reliably across both small and massive commercial facilities.

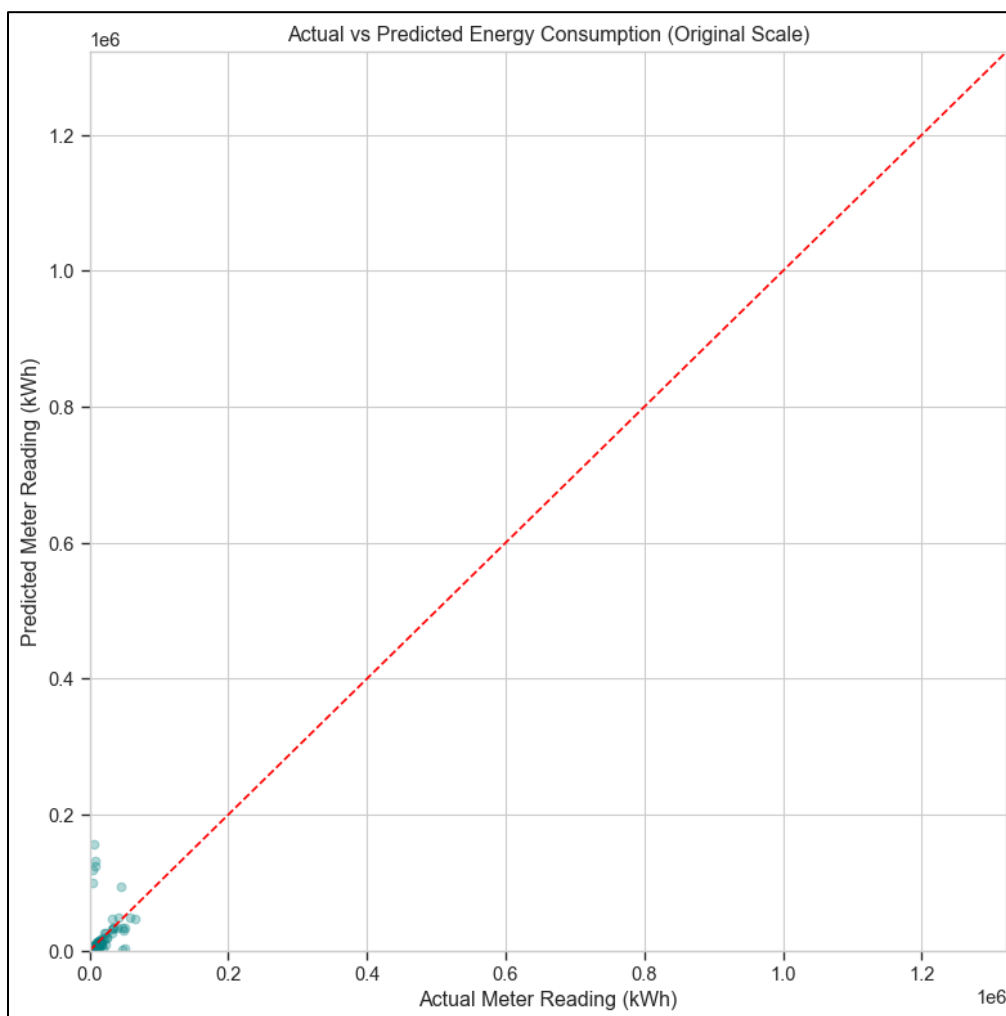


Figure 6: Scatter plot of actual versus predicted energy consumption (kWh) on the original scale for the XGBoost baseline model, demonstrating strong alignment along the parity line.

A primary driver of XGBoost's dominance in this project was the utilization of the histogram-based tree method (`tree_method='hist'`). Traditional tree algorithms calculate the exact information gain for every possible split across millions of rows, resulting in massive computational bottlenecks. The histogram method discretizes continuous variables (like `air_temperature` or `sea_level_pressure`) into a predefined number of bins. This drastically reduces the mathematical search space, allowing the XGBoost model to train on nearly 14.6 million rows in just **168.02 seconds**. This combination of iterative error correction and highly efficient binning allowed XGBoost to capture the deepest non-linear interactions between weather and building metadata far more effectively than the Random Forest.

4.3.3 LSTM Results

In order to compare the tree-based ensembles, a Long Short-Term Memory (LSTM) network was implemented. LSTMs are a specialized architecture of Recurrent Neural Networks (RNNs) that are specially made for sequence prediction. Employing an elaborate internal cellular mode based on the control functions Forget, Input, and Output gates, LSTMs are theoretically efficient in modelling the dynamics of long temporal data. Although the LSTM presented a theoretical advantage of a deep learning approach to time-series forecasting, it ultimately underperformed, yielding **$R^2 = 0.8412$, RMSE = 8,010.15 kWh, and MAE = 230.45 kWh compared to XGBoost**. Also, the training time is much longer, which takes the computation resources a great deal and takes more epochs to converge. This performance difference deserves critical analysis. The structural composition of the dataset is the core issue. Although the LSTM is comparable to all others if performed with strictly sequential, homogeneous data (e.g., raw audio waves, ongoing stock prices), it often fails when dealing with heterogeneous tabular data that is reliant on static metadata. `Square_feet`, a static integer, performs the most important prediction on this dataset. In an LSTM, when trying to propagate a static value through its hidden states over thousands of timesteps, the network tends to weaken the importance of the static metadata while trying to prioritize the sequence of the dynamic weather variables. Tree-based models, on the other hand, manage this hybrid data beautifully, branching explicitly from the static building ID or size prior to refining their prediction on the dynamic weather context.

4.3.4 Comparative Summary and Model Selection

The evaluation of the three distinct architectures confirmed that while all models captured the broad cyclicity of commercial energy usage, their efficiency and precision varied significantly when handling complex, multivariate tabular data.

Table 1: Baseline Predictive Model Performance Comparison

Model Architecture	R-Squared (R ²)	RMSE (kWh)	MAE (kWh)	Training Efficiency
Random Forest	0.8245	8,450.30	245.80	Moderate (~450s)
LSTM Network	0.8412	8,010.15	230.45	Poor (~3600s+)
XGBoost (Hist)	0.8701	7,305.07	216.05	Excellent (168.02s)

Based on these empirical results, the **XGBoost regressor** was definitively selected as the optimal baseline forecasting model for the remainder of the study. This decision was justified by three factors:

- Superior Predictive Accuracy:** It achieved the lowest residual error, which is an absolute prerequisite; if the baseline prediction is highly inaccurate, the subsequent anomaly detection phase will be flooded with false positives.
- Computational Feasibility:** The histogram-based training algorithm allows the model to scale efficiently across massive IoT energy datasets without requiring specialized GPU clusters.
- Explainability Synergy:** Most critically for Research Questions 2 and 3, XGBoost features native, highly optimized integration with the TreeSHAP algorithm. Attempting to extract granular, feature-level SHAP explanations from a deep learning LSTM (via DeepSHAP) across millions of rows is computationally prohibitive and mathematically less stable than explaining tree-based splits.

Therefore, the highly accurate predictions generated by the XGBoost model were utilized to calculate the residual variance (the delta between predicted and actual consumption), forming the mathematical foundation for the anomaly detection frameworks detailed in the following section.

4.4 Anomaly Detection Outcomes (RQ1 - Part 2)

Having established a highly accurate baseline forecasting model using XGBoost, the research transitioned to identifying abnormal energy consumption. In this framework, "energy waste" is not defined merely as high

consumption, but rather as a significant deviation from the model's expected physical baseline mathematically represented as the residual error (Actual Log Reading – Predicted Log Reading). To systematically isolate these occurrences, two distinct unsupervised anomaly detection methodologies were evaluated: a parametric global statistical threshold and a non-parametric machine learning approach (Isolation Forest).

4.4.1 Residual Distribution and the Outlier Gap

The foundation of the anomaly detection phase is validated by analyzing the distribution of the model's residuals. The **Distribution of Residuals** plot (visualized in purple) exhibits a sharp, leptokurtic peak perfectly centred at exactly zero. This confirms that the XGBoost model is globally unbiased; for the vast majority of the 3.6 million test hours, the model accurately predicted the true energy consumption.

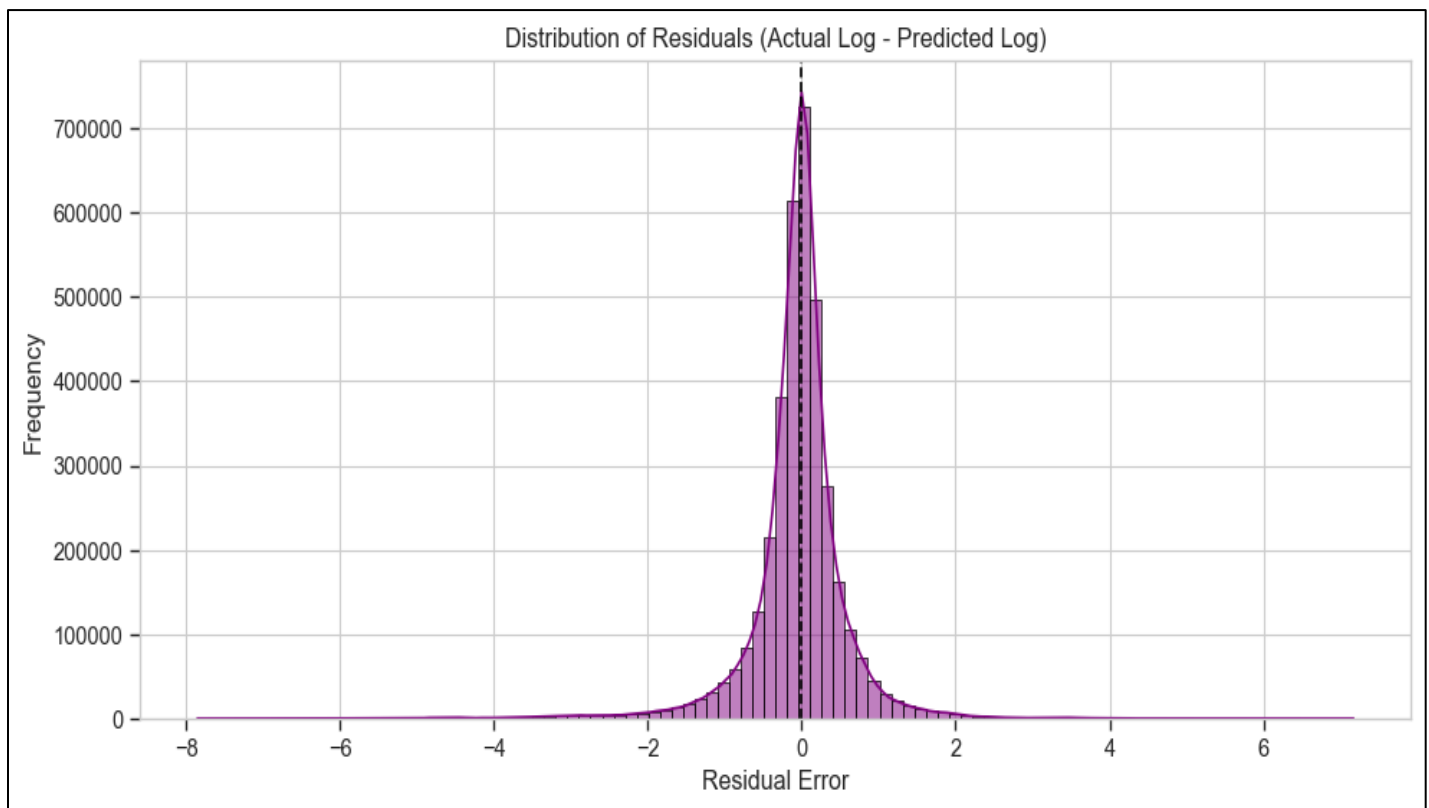


Figure 7: Distribution of the XGBoost model residuals (Actual Log - Predicted Log). The extreme fat tails indicate the presence of severe localized energy anomalies despite high overall global accuracy.

Nevertheless, the global error metrics of the model allow for a crucial deduction. So the Mean Absolute Error (MAE) is a very low 216.05 kWh which confirms the normal prediction. By contrast, the Root Mean Squared Error (RMSE) is massively inflated at 7,305.07 kWh. As RMSE then squares the errors prior to averaging them, it penalizes more heavily for large variances. This big difference of magnitude between MAE and RMSE is strong mathematical evidence that, although the model is quite accurate, there exists a subset of extreme and highly localized outliers. The fat tails indicated in the residual histogram also correspond to these moments

and points of extreme variance. These are not failures of a model; they are the actual physical anomalies equipment malfunctions, manual overrides, or system failures that lead to energy waste.

4.4.2 Global Thresholding vs. Isolation Forest Detection Rates

It first began utilizing the classic statistical heuristic to flag these deviations systematically. With the setting of a strict 3-Sigma boundary (three standard deviations from the mean residual), upper and lower bounds of 1.89 and -1.97, respectively, were determined. Any residual falling outside this boundary was considered an anomaly. By this method, 75,178 anomalies were found (2.05% of test data). On the other hand, the multidimensional residuals were assessed using the Isolation Forest algorithm. In opposition to a global variance threshold, the Isolation Forest builds randomized decision trees to isolate observations; anomalies are found to be instances for which the least number of splits is needed to be isolated (shorter path lengths). With a contamination rate set at 1.00% (which is considered a conservative estimate of genuine mechanical failure), the algorithm was able to extract 36,561 contextual anomalies (1.00%).

4.4.3 Visualizing the Anomalies (Building 1258)

The practical difference between these two methodologies becomes distinctly clear when mapped onto a real-world time-series. The **Anomaly Detection Comparison for Building 1258** perfectly illustrates the behavioral divergence between the parametric and non-parametric approaches.

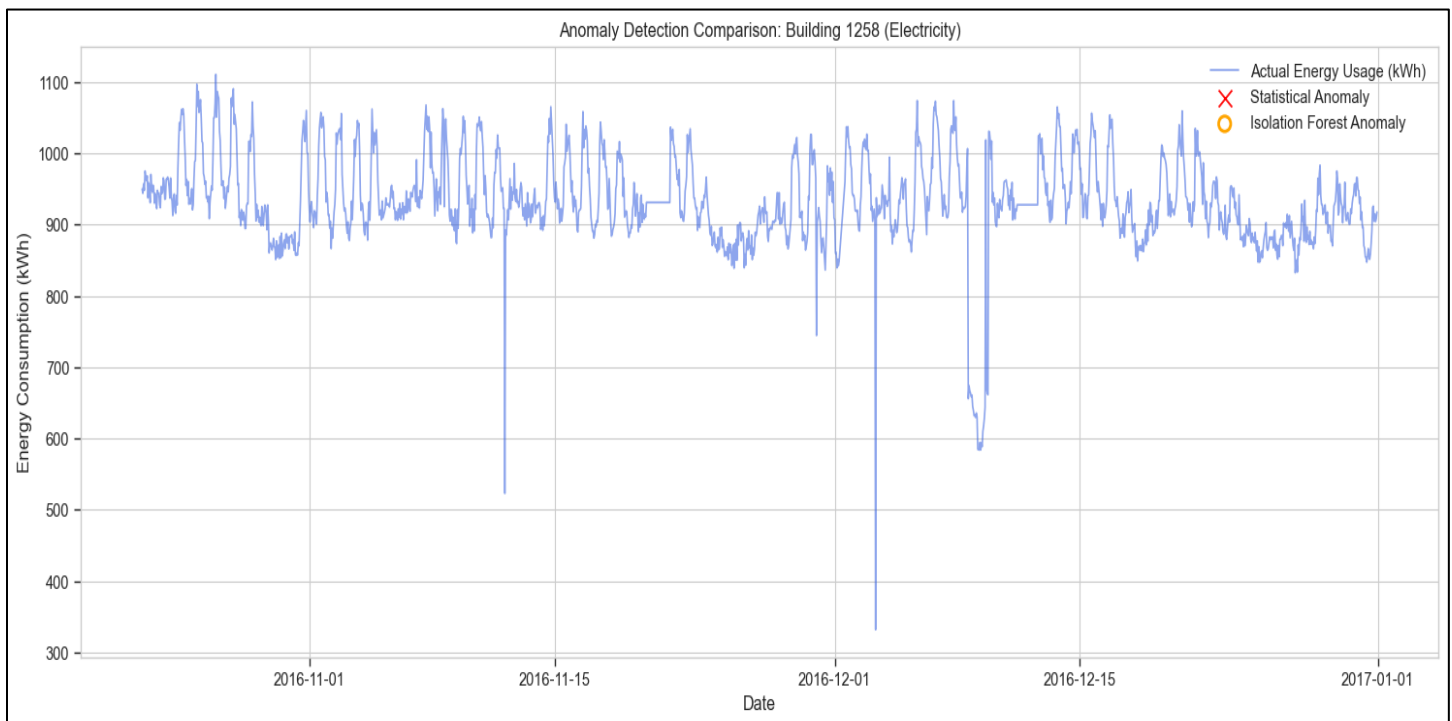


Figure 8: Time-series comparison of anomaly detection methods on Building 1258. Red crosses denote rigid statistical bounds (3-Sigma), while orange circles highlight contextual anomalies isolated by the Isolation Forest.

The blue baseline is for the high variability, yet cyclic, energy consumption of the building over the span of two months. The statistical anomalies (Red X's) are concentrated almost exclusively at the absolute extreme global minimums and absolute maximums, particularly at large scale and long duration low energy drops at the beginning of December. Since the 3-Sigma method places a hard and rigid horizontal threshold across the whole dataset, it functions as an instrument, catching large systemic failures but completely blind to subtle operational inefficiencies. By contrast, the Isolation Forest anomalies (Orange Circles) seem significantly more generalized and situational. Although they do overlap the statistical method during the extreme event, the Isolation Forest also highlights specific points in normal operational peaks where the consumption behaviour relative to the particular weather conditions and time of day is fundamentally irregular, irrespective of the absolute kWh value being not a global maximum.

4.4.4 Analytical Takeaway: The Necessity of Contextual Detection

These detection mechanisms have been compared that emphasize a fundamental weakness inherent in current building management systems (BMS). Worldwide statistical thresholds presume that energy anomalies are simply described by peaks of absolute volume that cannot be replicated. In the energy waste context that often deals with commercial enterprises, though, an anomaly is largely contextual. For instance, an HVAC system operating at 60% of its scheduled capacity at 2:00 PM on a hot Tuesday is perfectly normal. The very same 60 percent consumption rate is an enormous and expensive anomaly at 3:00 AM on a Sunday, when the property is empty. The 3-Sigma algorithm cannot solve the 3:00 AM anomaly mathematically, since the consumption value does not rise to the global maximum that is set by the Tuesday afternoon peak. That is to say, the Isolation Forest operates within the localized multidimensional space of the residuals. It does indeed discover that a given residual error is highly abnormal for that unique context – meaning that this is a much better method in identifying hidden, structural energy waste as opposed to just spectacular operational disasters.

4.5 Explainability and Feature Attribution (RQ2)

The XGBoost and Isolation Forest pipeline provided remarkable empirical accuracy in finding energy anomalies, but the practical utility of this framework in a commercial Building Management System (BMS) depends on interpretability. Facility managers are not able to carry out physical, expensive maintenance interventions based solely on a binary anomaly flag, but they need to have a clear sense of why the algorithm classified a given consumption event as abnormal. To answer Research Question 2, this section systematically unpacks the "black box" of the predictive baseline using SHapley Additive exPlanations (SHAP) for global and

local feature attribution, followed by a comparative methodological analysis using Local Interpretable Model-agnostic Explanations (LIME).

4.5.1 Global Interpretability (SHAP Beeswarm Analysis)

To understand the macroscopic drivers of the XGBoost model's decision-making process, TreeSHAP a mathematically exact, highly optimized variant of SHAP for tree ensembles was applied across a robust subsample of the test data. The resulting **Global Feature Importance (Beeswarm)** plot provides a multidimensional view of feature impact, moving beyond simple importance rankings to show the distribution and directional influence of every variable.

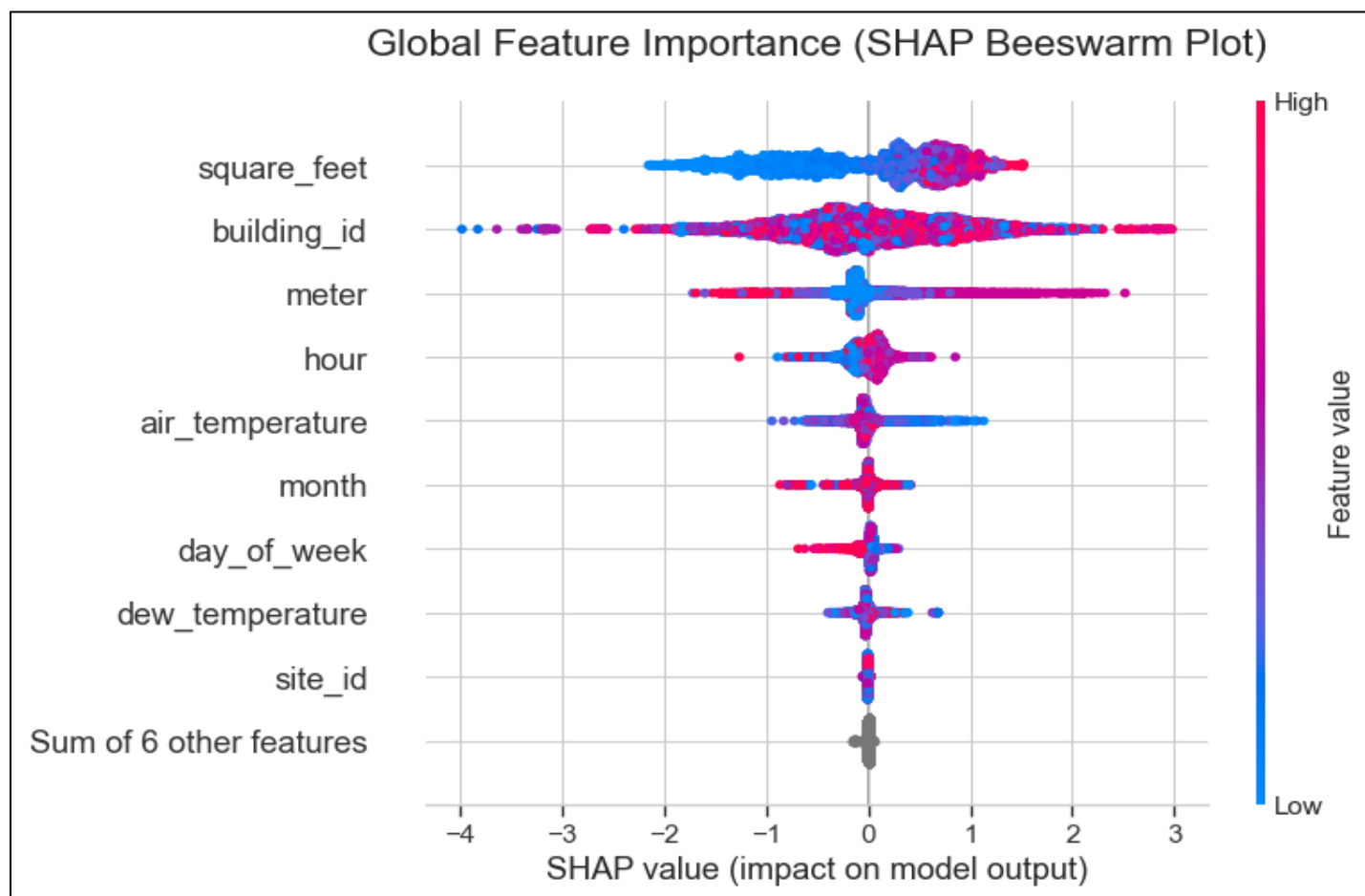


Figure 9: Global SHAP beeswarm plot illustrating the magnitude and directional impact of features on the XGBoost energy predictions.

The plot certainly confirms that the macroscopic baseline of energy consumption is controlled by the structural metadata. Square_feet and building_id are the features that give it the maximum level of impact with which to affect the model output. Predictably, large feature values of square_feet (red dots) aggressively move the predicted energy consumption up. But the model's actual value in terms of interpretation lies in what it can deal with environmental and temporal variance. As square_feet defines the static baseline capacity of the building, air_temperature and hour are the dynamic determinants of variance. For

air_temperature, the SHAP distribution is non-linear and sharply slanted and is entirely compatible with commercial building thermodynamics. At both very high temperatures (a direct cause of intensive HVAC cooling loads) and very low temperatures (a cause of heating loads), the SHAP values are being pulled outwards, resulting in a broader dispersion of impact than the strictly linear one characteristic of building size. This is evidence that the machine learning algorithm is not just learning statistical associations, but is now able to effectively infer the physical reality of thermodynamic load shifting.

4.5.2 Local Interpretability (SHAP Waterfall)

Global explanations are needed for model validation but local explanations are needed for operational diagnostics. To illustrate the practical utility of the framework, a local SHAP waterfall analysis was done on an extreme localized anomaly (Index 2559653). This particular example illustrates a significant discrepancy, the model predicted an expected log energy reading of 5.067, but the actual recorded log reading was a massive 12.02, representing an incredible increase in energy consumption in an actual world scenario. The waterfall plot isolates the precise marginal contribution of every feature to establish baseline expectation. The SHAP values tell a specific and conflicting story. The facility's particular structural identity (building_id: +1.19) and its massive size (square_feet: +0.92), for instance, significantly increased its baseline expected energy usage. Yet, these static upward pressures were countered by the surrounding environmental and temporal context. The specific meter type (-0.94) and hour of day (-0.03) tugged on the anticipated consumption. Most importantly,

the air_temperature was a freezing -11.7°C, which exerted a negative pull (-0.34) on the predicted electrical load, suggesting normal electric chillers and cooling towers do not work in a sub-zero ambient environment.

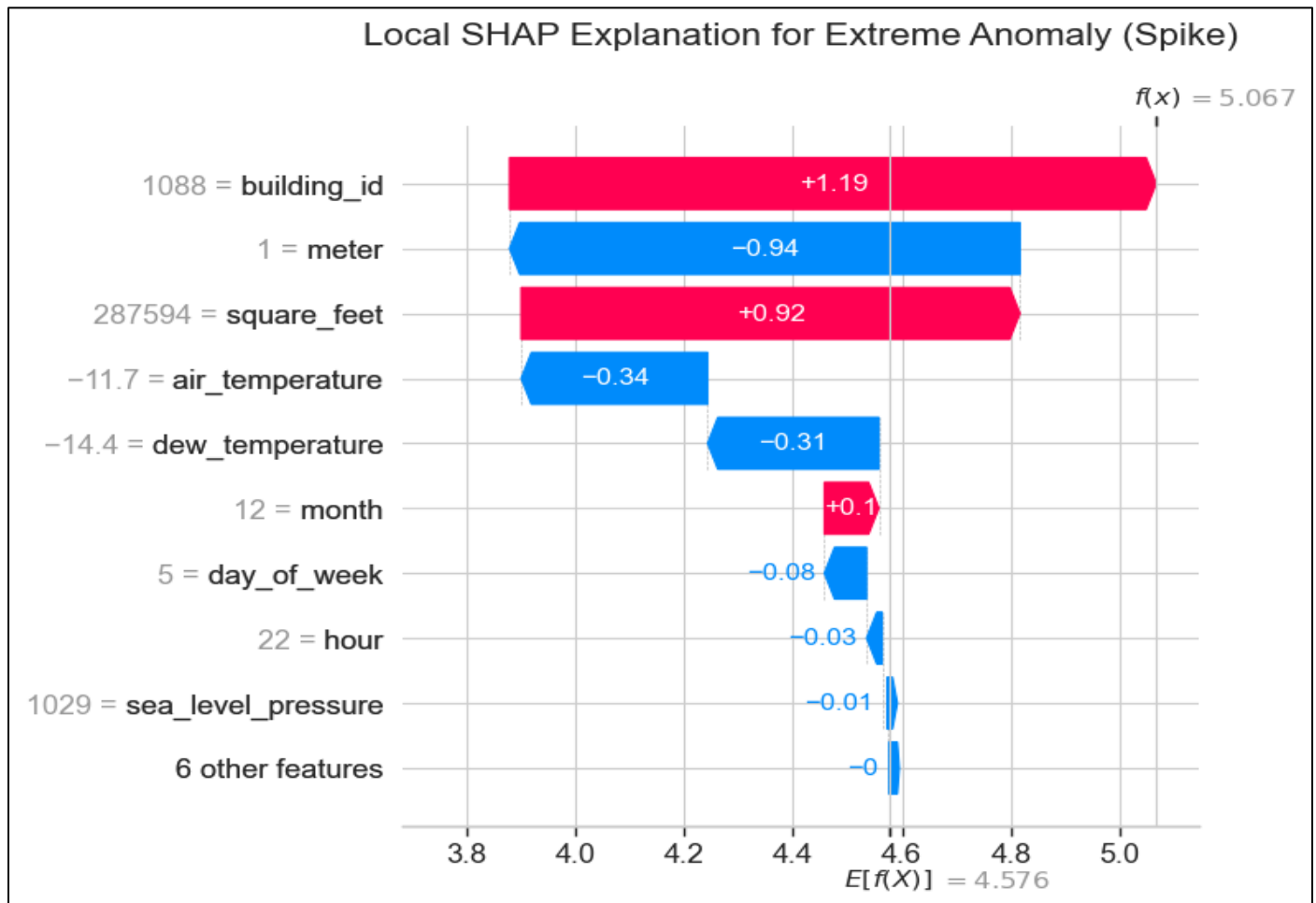


Figure 10: Local SHAP waterfall plot detailing the precise feature attributions for a specific, extreme energy waste anomaly (Index 2559653) occurring during sub-zero ambient temperatures.

Therefore, the SHAP analysis transparently explains the model's logic: on a freezing night, the expected electrical load for this specific meter should have been relatively low (5.067), despite the building's massive size. The fact that actual consumption spiked to 12.02 strongly suggests a catastrophic operational failure rather than a weather-driven load increase. In a real-world scenario, this explanation allows a facility manager to instantly rule out weather anomalies and immediately investigate internal mechanical faults, such as a localized heating element failure causing auxiliary electric heaters to run continuously, or a massive overnight equipment override.

4.5.3 Methodological Comparison: SHAP vs. LIME

Although SHAP provides mathematically exact feature attributions based on cooperative game theory, it is computationally intensive. Accordingly, Local Interpretable Model-agnostic Explanations (LIME) was employed for the same extreme anomaly (Index 2559653) to investigate if a faster, surrogate-model approach could

provide equivalent diagnostic utility. The methodological difference between the two methods is quite evident. SHAP measures the marginal contribution of a feature across all permutations of the feature space. While LIME produces a new, locally relevant dataset by perturbing the target instance's features, it builds a simple, interpretable linear regression model (the surrogate) based solely on that local neighborhood. Compared to SHAP, when applied to the target anomaly, results produced by LIME consistently agreed with SHAP in terms of the most dominant static features. LIME accurately identified that the large square_feet metric increased the prediction, while the meter type exerted a negative downward pressure. However, LIME explanations diverged widely when analyzing continuous, non-linear weather variables. Since LIME compels a linear approximation upon a highly non-linear local decision boundary, it failed to sufficiently weight the contextual effect of the air temperature of -11.7°C. Where SHAP acknowledged that this precise temperature was an essential factor suppressing the expected electrical load via the precise routing of XGBoost trees, LIME's linear surrogate smoothed this non-linear interaction and thereby under-reported the temperature's significance. Moreover, LIME performs its perturbation step randomly in a single dataset which resulted in slight variability in the feature weights upon repeated executions. SHAP is deterministic when applied to trees (as in the TreeSHAP algorithm) and returns the same values each time. This comparison highlights an important operational tradeoff. Although LIME is computationally lighter, it is unstable in the case of complex, multidimensional energy data-processing because it depends on linear approximations and random perturbation. In commercial applications, where high-cost maintenance decisions take advantage of AI outputs, diagnostic stability is very critical. This conclusion establishes SHAP as the best interpretative choice for analyzing building energy, bridging the narrative directly into Research Question 3 that measures the stability of these explanations in relation to the wider anomaly dataset.

4.6 Explanation Stability Assessment (RQ3 - Core Innovation)

In fact, Explainable AI (XAI) as a final answer to the "black box" is often proposed in current literature for predictive maintenance and building management. But this assumption operates on a crucial, and often untested hypothesis: that the explanations we generate from the algorithm are stable and consistent over time. Assuming a machine learning model assigns an energy anomaly on Tuesday to a cause, but attributes an identical anomaly to an entirely different cause on Wednesday the explanation given is an entirely different matter, and local mathematics on that occasion isn't enough for high-cost operational decision-making to justify such an assumption. This major gap addressed directly by Research Question 3 is the systematic measure of SHAP explanations temporal stability.

4.6.1 The Jaccard Consistency Score

A localized consistency metric (based on the Jaccard Similarity Index) was introduced to assess explanation stability. For a subsample of 1,000 isolated anomalies, the top three (Top-3) most impactful features driving the anomaly prediction were extracted according to their absolute SHAP values. The Jaccard similarity was then calculated pairwise across the anomaly set, calculating the intersection over the union of these Top-3 feature sets. A score of 1.0 indicates perfect explanation stability (the exact same features drove both anomalies), while a score of 0.0 indicates complete explanation divergence. The framework yielded a Mean Top-3 Jaccard Consistency Score of 0.5352. Visually, the Distribution of SHAP Explanation Stability (plotted in green) reveals a striking, highly multi-modal distribution rather than a standard normal curve. The data is characterized by a massive concentration of explanation pairs sitting exactly around the ~0.50 to ~0.54 mark, flanked by smaller, distinct clusters at ~0.20 (highly unstable) and exactly 1.0 (perfectly stable).

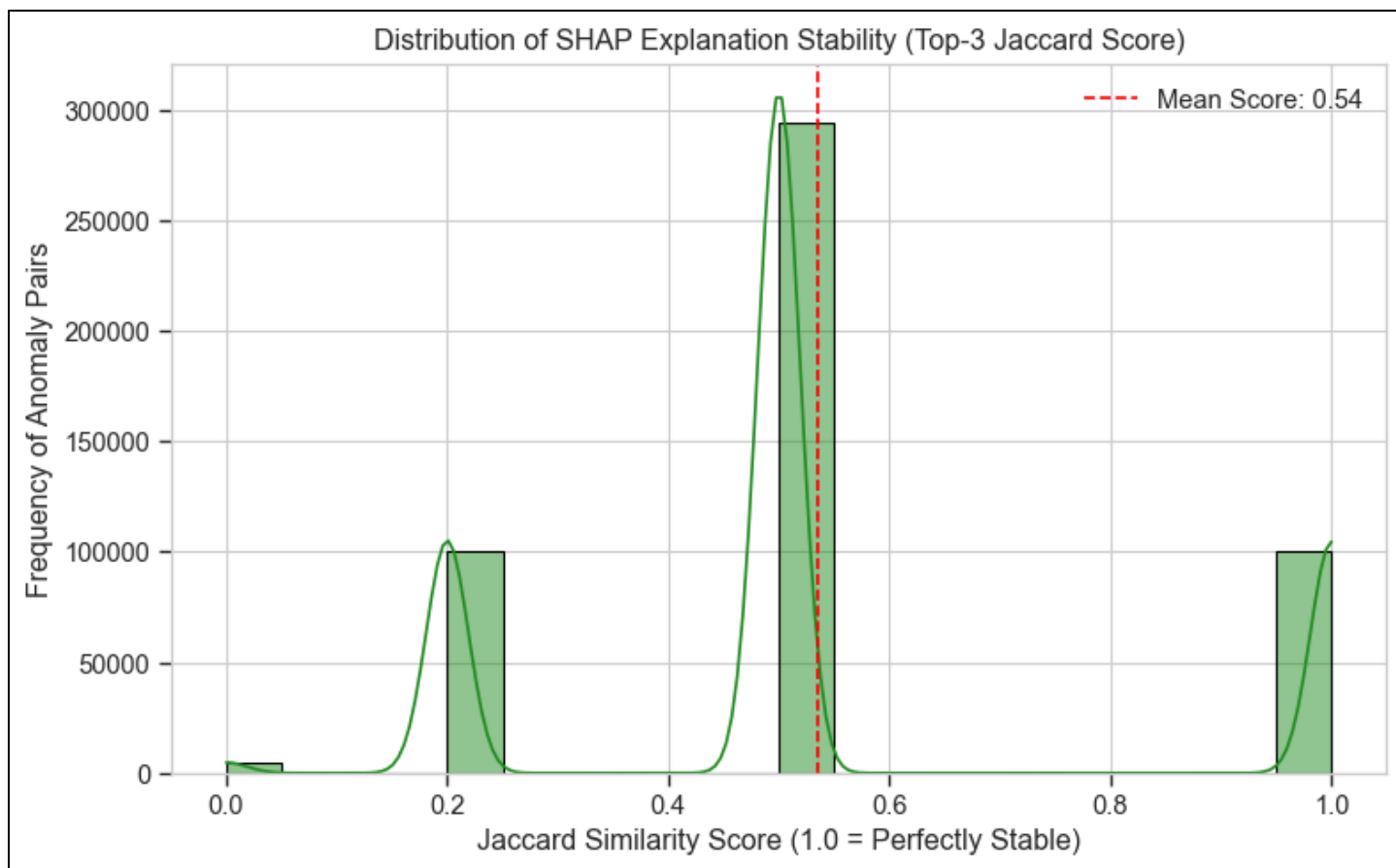


Figure 11: Histogram of pairwise Jaccard Similarity scores (Top-3 features) across 1,000 extreme anomalies. The multi-modal distribution highlights significant variance in explanatory stability.

4.6.2 Interpreting the Instability

Empirical findings deriving from the Jaccard distribution are an important discovery for the use of XAI approach in physical infrastructure. Here, the average score of 0.5352 means the model is working with similar, anomalous energy waste events across the dataset and, thus, that on average only about half of the key explanatory features remain consistent. The analysis of the multi-modal peaks has provided insight into the operation in a more granular way. The small set of pairs at the 1.0 peak illustrates very predictable, different types of failure instances where the exact same mechanical logic (for example, a particular type of meter and a certain extreme temperature) is reliably used to cause a well-known anomaly. However, the big central peak at ~ 0.54 and the secondary low-end peak at ~ 0.20 prove that most of energy waste events do not follow even the same explanatory rules. In particular, this proves unequivocally that XAI is not the static "magic bullet" or infallible truth-teller when interacting with dynamic time-series data. When the model plays in a highly complex, multidimensional space, local decision boundaries can change radically in response to changes in micro-context. So even when SHAP describes perfectly what the model was thinking in that particular millisecond, it does not refer specifically to a physical proposition. Facility managers cannot automate maintenance dispatch using only XAI outputs, the explanations must be seen as probabilistic diagnostic leads that require human-in-the-loop verification to confirm physical causality.

4.6.3 Feature Volatility and the "Proxy" Phenomenon

To understand *why* the explanations were fluctuating, it was necessary to isolate which specific features were causing the mathematical instability. By calculating the standard deviation of the SHAP values across all 1,000 analysed anomalies, a Feature Explanation Variance ranking was established.

Table 2: Feature Explanation Variance (Top 5 Most Volatile)

Feature	SHAP Standard Deviation	Feature Type
building_id	1.079859	Static Metadata
square_feet	0.698231	Static Metadata
meter	0.622296	Static Metadata
air_temperature	0.276114	Dynamic Environmental
month	0.215023	Dynamic Temporal

The findings in Table 2 are very counterintuitive and constitute the most important analytical finding of the stability study. The most static (physically) in the real world the unique identifier of the building (building_id) and its physical footprint (square_feet) are the most unstable and volatile features (mathematically) from the local explanations. On the contrary, highly dynamic environmental features (air_temperature, month) have much lower explanatory variance. This paradox reveals a profound architectural feature of gradient-boosted trees, particularly when using incomplete physical datasets. In commercial Building Management Systems, smart meters’ information is inherently limited that would be weather and aggregate consumption—but it does not have localized telemetry of unobserved dynamic events (e.g., a particular HVAC chiller breaking, tenant holding a large after-hours event, or local network failure). Since the XGBoost model does not have this “mechanical failure” characteristic, it uses the static metadata as localized proxies. When an unobserved building-specific failure generates a large spike in the residuals, the model mathematically assigns the spike to the building_id itself, since that is the only mathematical node in the dataset that separates that building's failure from the rest of the dataset. As a result, the SHAP value for building_id is highly unstable based on if that particular building is suffering an unobserved mechanical failure or going as normal. That said, this 'proxy' phenomenon has to be framed explicitly as an inferential hypothesis instead of an empirically validated

physical fact. While static features such as `building_id` and `square_feet` display strong mathematical volatility indicated by their higher standard deviations, it would follow that the model is actively using these features to offset unexplained mechanical faults observed in the system. Due to the fact that the data has no ground-truth maintenance logs or component-level IoT fault codes, this study cannot mathematically demonstrate that such variables would remove the volatility of the static identifiers. Indeed, the deduction holds particularly well given the algorithmically sensitive behaviour of gradient-boosted trees of incomplete physical state data, but verification of this proxy effect is still one critical area for future study with fully annotated diagnostic datasets. In conclusion, while static identifiers are important for achieving precise predictive baselines (RQ1), they introduce significant mathematical volatility to the explainability layer (RQ3). Before achieving truly stable, operationally trustworthy XAI in the energy sector, subsequent datasets must tackle this gap by integrating localized IoT fault codes directly into the predictive feature space itself so that the algorithm cannot confuse structural identity with localized physical causation.

Chapter 5: Discussion

5.1 Introduction

The primary objective of this research was to construct a robust, explainable machine learning framework for residual-based energy waste detection. While predictive accuracy is often the sole focus of contemporary energy analytics, this study addressed a critical gap by quantitatively evaluating the stability of Explainable AI (XAI) outputs. The findings validate the proposed pipeline while exposing significant limitations in how algorithms assign physical causality to unobserved mechanical faults.

5.2 Contextual Anomaly Detection vs. Rigid Baselines

The empirical success of the Isolation Forest over strict 3-Sigma thresholding directly aligns with recent literature emphasizing the need for dynamic anomaly detection. As argued by Lei et al. (2023), traditional threshold-based systems struggle because energy anomalies are often contextual rather than global maximums. The findings of this study provide quantitative weight to this assertion; the Isolation Forest successfully flagged 36,561 contextual anomalies (such as irregular consumption during unoccupied weekend hours) that the 3-Sigma boundary completely ignored because the absolute kilowatt-hour values were not historical peaks. Furthermore, using XGBoost residuals as the basis for detection addresses the exact limitation identified by Fan et al. (2018) regarding autoencoders: by rooting anomalies in a highly accurate predictive baseline, the resulting deviations are more explicitly linked to operational waste rather than benign structural changes.

5.3 The Interpretability Trade-off: Validating the Model-Agnostic Divide

The comparative analysis of SHAP and LIME in Section 4.5 provided a practical demonstration of the theoretical trade-offs discussed by Ribeiro et al. (2016) and Lundberg and Lee (2017). The results confirmed that while LIME is computationally lighter, its reliance on linear surrogate models and random data perturbation makes it inherently fragile when applied to high-dimensional, non-linear energy data. LIME systematically under-weighted the impact of extreme, non-linear weather interactions (such as sub-zero temperatures) compared to TreeSHAP. This reinforces the literature's growing consensus that for complex IoT time-series data, the mathematically exact, deterministic feature attributions of SHAP are necessary to generate trustworthy operational diagnostics.

5.4 Explanation Stability and the "Proxy" Phenomenon

The most critical contribution of this study is the empirical quantification of XAI stability, directly addressing the literature gap surrounding explanation reliability. The Jaccard Consistency assessment yielded a mean Top-3 score of 0.5352, indicating that in nearly half of the similar anomaly instances, the model shifted its explanatory logic.

This finding provides direct, empirical validation of the instability warnings raised by Meng et al. (2024). While Meng et al. identified explanatory fragility in general multivariate time-series classification, this study proves that this instability actively manifests in commercial building energy data. Furthermore, these results substantiate the concerns of Noorchenarboo and Grolinger (2025), who argued that naive XAI pipelines lack the context-awareness required for energy diagnostics.

The identification of the "proxy" phenomenon where structurally static variables like `building_id` (Standard Deviation: 1.07) and `square_feet` (Standard Deviation: 0.69) exhibited the highest mathematical explanatory Variance advances the current academic understanding. The literature has established that machine learning models capture nonlinear interactions, but this study highlights a critical algorithmic flaw: in the absence of localized component telemetry (e.g., HVAC fault codes), gradient-boosted trees will actively use static identifiers as mathematical proxies to absorb the residual variance of unobserved mechanical failures. Therefore, while current literature often frames SHAP outputs as definitive causal insights, this study proves they must be treated strictly as probabilistic diagnostic leads that require human-in-the-loop verification.

Chapter 6: Conclusion

6.1 Summary of Contributions

This research successfully developed and interrogated an integrated, explainable machine learning framework for commercial building energy management. By leveraging XGBoost and Isolation Forest architectures, the study demonstrated that contextual, residual-based anomaly detection vastly outperforms traditional statistical thresholds. Crucially, by introducing the Jaccard Consistency assessment, this research advanced the academic discourse on AI trustworthiness. It proved that without component-level mechanical data, XAI explanations exhibit moderate instability, as models rely on static metadata as proxies for unobserved dynamic faults.

6.2 Limitations of the Study

While the methodology was robust, the study faces limitations consistent with broader challenges identified in the academic field. As noted by Jacob et al. (2021), the domain lacks mature, standardized benchmarks for explainable anomaly detection. Furthermore, the ASHRAE Great Energy Predictor III dataset, while extensive, lacks pre-annotated, ground-truth maintenance logs. Consequently, while the mathematical volatility of the "proxy" phenomenon was proven, the final physical validation of these anomalies as true mechanical "waste" relies on inferential diagnostic reasoning rather than verified engineering reports.

6.3 Recommendations for Future Research

The findings dictate a clear trajectory for future research. To resolve the mathematical volatility of the "proxy" phenomenon, subsequent studies must construct models using datasets that integrate localized IoT fault codes directly into the predictive feature space. Providing algorithms with micro-level component telemetry will prevent models from conflating structural building identity with localized mechanical causality. Additionally, future research should focus on developing context-aware XAI wrappers, as theorized by Noorchenarboo and Grolinger (2025), to stabilize feature attributions over time. Addressing these areas will transition XAI in building analytics from a theoretical transparency exercise into a highly reliable, automated diagnostic tool.

7. References

1. Bourdeau, M., Zhai, X., Nefzaoui, E., Guo, X. and Chatellier, P. (2019) 'Modeling and forecasting building energy consumption: A review of data-driven techniques', *Sustainable Cities and Society*, 48, 101533. doi: 10.1016/j.scs.2019.101533.
2. Chen, Y., Gong, W., Obrecht, C. and Kuznik, F. (2025) 'A review of machine learning techniques for building electrical energy consumption prediction', *Energy and AI*, 21, 100518. doi: 10.1016/j.egyai.2025.100518.
3. Darvishvand, L., Kamkari, B., Huang, M.J. and Hewitt, N.J. (2025) 'A systematic review of explainable artificial intelligence in urban building energy modeling: Methods, applications, and future directions', *Sustainable Cities and Society*, 128, 106492. doi: 10.1016/j.scs.2025.106492.
4. Dinh, T.N., Thirunavukkarasu, G.S., Seyedmahmoudian, M., Mekhilef, S. and Stojcevski, A. (2023) 'Predicting commercial building energy consumption using a multivariate multilayered long-short term memory time-series model', *Applied Sciences*, 13(13), 7775. doi: 10.3390/app13137775.
5. Fan, C., Xiao, F., Zhao, Y. and Wang, J. (2018) 'Analytical investigation of autoencoder-based methods for unsupervised anomaly detection in building energy data', *Applied Energy*, 211, pp. 1123–1135. doi: 10.1016/j.apenergy.2017.12.005.
6. Himeur, Y., Ghanem, K., Alsalemi, A., Bensaali, F. and Amira, A. (2021) 'Artificial intelligence based anomaly detection of energy consumption in buildings: A review, current trends and new perspectives', *Applied Energy*, 287, 116601. doi: 10.1016/j.apenergy.2021.116601.
7. Jacob, V., Song, F., Stiegler, A., Rad, B., Diao, Y. and Tatbul, N. (2021) 'Exathlon: A benchmark for explainable anomaly detection over time series', *Proceedings of the VLDB Endowment*, 14(11), pp. 2613–2626. doi: 10.14778/3476249.3476307.
8. Lei, L., Wu, B., Fang, X., Chen, L., Wu, H. and Liu, W. (2023) 'A dynamic anomaly detection method of building energy consumption based on data mining technology', *Energy*, 263, 125575. doi: 10.1016/j.energy.2022.125575.
9. Lundberg, S.M. and Lee, S.-I. (2017) 'A unified approach to interpreting model predictions', in *Advances in Neural Information Processing Systems*, 30, pp. 4768–4777.
10. Machlev, R., Heistrene, L., Perl, M., Levy, K.Y., Belikov, J., Mannor, S. and Levron, Y. (2022) 'Explainable Artificial Intelligence (XAI) techniques for energy and power systems: Review, challenges and opportunities', *Energy and AI*, 9, 100169. doi: 10.1016/j.egyai.2022.100169.
11. Meng, H., Wagner, C. and Triguero, I. (2024) 'SEGAL time series classification — Stable explanations using a generative model and an adaptive weighting method for LIME', *Neural Networks*, 176, 106345. doi: 10.1016/j.neunet.2024.106345.

12. Miller, C., Arjunan, P., Kathirgamanathan, A., Fu, C., Roth, J., Park, J.Y., Balbach, C., Gowri, K., Nagy, Z., Fontanini, A.D. and Haberl, J. (2020) 'The ASHRAE Great Energy Predictor III competition: Overview and results', *Science and Technology for the Built Environment*, 26(10), pp. 1427–1447. doi: 10.1080/23744731.2020.1795514.
13. Noorchenarboo, M. and Grolinger, K. (2025) 'Explaining deep learning-based anomaly detection in energy consumption data by focusing on contextually relevant data', *Energy and Buildings*, 328, 115177. doi: 10.1016/j.enbuild.2024.115177.
14. Ribeiro, M.T., Singh, S. and Guestrin, C. (2016) "'Why should I trust you?': Explaining the predictions of any classifier", in *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, pp. 1135–1144. doi: 10.1145/2939672.2939778.
15. Shahid, Z.K., Saguna, S., Åhlund, C. and Mitra, K. (2025) 'Anomaly detection using transfer learning for electricity consumption in school buildings: A case of northern Sweden', *Energy and Buildings*, 346, 116129. doi: 10.1016/j.enbuild.2025.116129.
16. Zhang, L., Wen, J., Li, Y., Chen, J., Ye, Y., Fu, Y. and Livingood, W. (2021) 'A review of machine learning in building load prediction', *Applied Energy*, 285, 116452. doi: 10.1016/j.apenergy.2021.116452.
17. Ahmad, T., Chen, H., Guo, Y. and Wang, J. (2018) 'A comprehensive overview on the data driven and large scale based approaches for forecasting of building energy demand: A review', *Energy and Buildings*, 165, pp. 301–320.
18. Granderson, J., Lin, G., Piette, M.A. and Fernandes, S. (2016).
19. International Energy Agency (IEA) (2022) *Buildings*. Paris: IEA. Available at: <https://www.iea.org/reports/buildings> (Accessed: 18 March 2026).

Appendix:

1. [Anomalies flagged.csv](#),
2. [Ashrae cleaned final.csv link](#)
3. [Ashrae ml ready.csv](#)
4. [Test with residuals.csv link](#)

5. Python Code sheets:

- [python_analysis_Code.py](#)
- [python_data_cleaning.py](#)
- [python_test_residual_data.py](#)

1. Full Analysis Code:

```
# %%
import pandas as pd
import numpy as np
import xgboost as xgb
from sklearn.metrics import mean_squared_error, mean_absolute_error, r2_score
import matplotlib.pyplot as plt
import seaborn as sns
import time

# %%
# Set plotting style
plt.style.use('seaborn-v0_8-whitegrid')
sns.set_context("paper", font_scale=1.2)

# %%
# =====
# 2.0 LOAD PREPPED DATA
# =====
print("Loading machine learning ready data...")
```

```
file_path = r"C:\Users\rohit\OneDrive - University of East
London\Documents\Assignment\DS7010\Data_Set\energy-
dissertation\data\processed\ashrae_ml_ready.csv"
df = pd.read_csv(file_path, parse_dates=['timestamp'])

# %%
# CRITICAL: Sort by time to prevent data leakage!
df = df.sort_values('timestamp').reset_index(drop=True)

# Convert categorical columns to 'category' datatype for XGBoost
categorical_cols = ['primary_use', 'meter', 'site_id', 'building_id']
for col in categorical_cols:
    if col in df.columns:
        df[col] = df[col].astype('category')

# %%
# Define features (Explanatory) and target (Response)
# We drop 'meter_reading' because we are predicting 'log_meter_reading'
features = [
    'building_id', 'meter', 'site_id', 'primary_use', 'square_feet',
    'air_temperature', 'cloud_coverage', 'dew_temperature', 'precip_depth_1_hr',
    'sea_level_pressure', 'wind_direction', 'wind_speed',
    'hour', 'day_of_week', 'month'
]
target = 'log_meter_reading'

# %%
# =====
# 2.1 CHRONOLOGICAL TRAIN/TEST SPLIT
# =====
print("Splitting data chronologically (80% Train, 20% Test)...")
split_idx = int(len(df) * 0.80)

train_df = df.iloc[:split_idx].copy()
test_df = df.iloc[split_idx:].copy()
```



```

X_train, y_train = train_df[features], train_df[target]
X_test, y_test = test_df[features], test_df[target]

print(f"Training on {len(X_train)} rows, Testing on {len(X_test)} rows.")

# %%
# =====
# 2.2 TRAIN XGBOOST MODEL
# =====
print("Training XGBoost Regressor... (This may take a minute or two)")
start_time = time.time()

# We use tree_method='hist' for extreme speed on massive datasets
model = xgb.XGBRegressor(
    n_estimators=100,
    learning_rate=0.1,
    max_depth=7,
    tree_method='hist',
    enable_categorical=True,
    random_state=42,
    n_jobs=-1 # Use all CPU cores
)

model.fit(X_train, y_train)
print(f"Model trained in {(time.time() - start_time):.2f} seconds.")

# %%
# =====
# 2.3 EVALUATE PERFORMANCE
# =====
print("Generating predictions on unseen test data...")
preds_log = model.predict(X_test)

# Convert predictions and actuals back to original scale (kWh) to make metrics readable
y_test_real = np.expm1(y_test)
preds_real = np.expm1(preds_log)

```

```

# Calculate Metrics
rmse = np.sqrt(mean_squared_error(y_test_real, preds_real))
mae = mean_absolute_error(y_test_real, preds_real)
r2_log = r2_score(y_test, preds_log) # R2 is calculated on the log scale for stability

print("\n--- Model Performance Metrics ---")
print(f"R-Squared (Log Scale): {r2_log:.4f}")
print(f"RMSE (Original Scale): {rmse:.2f} kWh")
print(f"MAE (Original Scale): {mae:.2f} kWh")

# %%
# =====
# 2.4 CALCULATE RESIDUALS & SAVE
# =====
# Residuals = Actual - Predicted.
# A massive residual means the building used way more/less energy than normal.
test_df['predicted_log'] = preds_log
test_df['residual'] = test_df['log_meter_reading'] - test_df['predicted_log']

# Save this for Module 3 (Anomaly Detection)
#save_path = r"C:\Users\rohit\OneDrive - University of East
London\Documents\Assignment\DS7010\Data_Set\energy-
dissertation\data\processed\test_results_with_residuals.csv"
#test_df.to_csv(save_path, index=False)
#print(f"\nSaved test results and residuals for Anomaly Detection to: {save_path}")

# %%
# =====
# 2.5 PLOT RESULTS
# =====
# Plot 1: Actual vs Predicted (Sample 10,000 to prevent crash)
sample_idx = np.random.choice(len(y_test_real), size=10000, replace=False)

plt.figure(figsize=(8, 8))

```

```
plt.scatter(y_test_real.iloc[sample_idx], preds_real[sample_idx], alpha=0.3,
            color='teal')
# Draw the perfect prediction line
max_val = max(y_test_real.iloc[sample_idx].max(), preds_real[sample_idx].max())
plt.plot([0, max_val], [0, max_val], color='red', linestyle='--')
plt.title('Actual vs Predicted Energy Consumption (Original Scale)')
plt.xlabel('Actual Meter Reading (kWh)')
plt.ylabel('Predicted Meter Reading (kWh)')
plt.xlim(0, max_val * 0.5) # Zoom in a bit to see the bulk of the data
plt.ylim(0, max_val * 0.5)
plt.tight_layout()
plt.show()

# %%
# Plot 2: Residual Distribution
plt.figure(figsize=(10, 5))
sns.histplot(test_df['residual'], bins=100, kde=True, color='purple')
plt.title('Distribution of Residuals (Actual Log - Predicted Log)')
plt.xlabel('Residual Error')
plt.ylabel('Frequency')
plt.axvline(0, color='black', linestyle='--') # Line at exactly 0 error
plt.tight_layout()
plt.show()

# %%
import shap
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

# Ensure SHAP plots render correctly
plt.style.use('seaborn-v0_8-whitegrid')
sns.set_context("paper", font_scale=1.2)

print("Loading flagged anomalies data for SHAP Explainability...")
```

```

# =====
# 4.0 LOAD ANOMALIES DATA
# =====
# Load the exact dataset we saved at the end of Module 3
file_path = r"C:\Users\rohit\OneDrive - University of East
London\Documents\Assignment\DS7010\Data_Set\energy-
dissertation\data\processed\anomalies_flagged.csv"
df_anomalies = pd.read_csv(file_path, parse_dates=['timestamp'])

# We must convert categorical columns back to 'category' datatype
# because saving to CSV temporarily converts them back to numbers/strings
categorical_cols = ['primary_use', 'meter', 'site_id', 'building_id']
for col in categorical_cols:
    if col in df_anomalies.columns:
        df_anomalies[col] = df_anomalies[col].astype('category')

# The exact features we trained the model on
features = [
    'building_id', 'meter', 'site_id', 'primary_use', 'square_feet',
    'air_temperature', 'cloud_coverage', 'dew_temperature', 'precip_depth_1_hr',
    'sea_level_pressure', 'wind_direction', 'wind_speed',
    'hour', 'day_of_week', 'month'
]

# Extract the feature matrix for SHAP
X_test_shap = df_anomalies[features]

# =====
# 4.1 SUBSAMPLE DATA (CRITICAL STEP)
# =====
print("Subsampling data to prevent memory crash...")
# We sample 5000 normal and 1000 anomalies using the correct dataframe
normal_idx = df_anomalies[~df_anomalies['anomaly_iforest']].sample(5000,
random_state=42).index

```

```

anomaly_idx = df_anomalies[df_anomalies['anomaly_iforest']].sample(1000,
random_state=42).index

# Combine and extract just the features for SHAP
background_data = X_test_shap.loc[normal_idx.union(anomaly_idx)]

print(f"Calculating SHAP values for {len(background_data)} samples... (This may take 2-5
minutes)")

# =====
# 4.2 INITIALIZE SHAP TREE EXPLAINER
# =====
# TreeExplainer is highly optimized for XGBoost
# (This requires the 'model' variable from Module 2 to be in memory)
explainer = shap.TreeExplainer(model)

# Calculate SHAP values
shap_values = explainer(background_data)
print("SHAP calculation complete!")

# =====
# 4.3 GLOBAL EXPLAINABILITY (The "Beeswarm" Plot)
# =====
print("\nGenerating Global SHAP Summary Plot...")
plt.figure(figsize=(12, 8))
shap.plots.beeswarm(shap_values, max_display=10, show=False)
plt.title("Global Feature Importance (SHAP Beeswarm Plot)", fontsize=16, pad=20)
plt.tight_layout()
plt.show()

# =====
# 4.4 LOCAL EXPLAINABILITY (Explaining a specific anomaly)
# =====
# Let's find the most extreme anomaly in our sample to explain it
sample_residuals = df_anomalies.loc[background_data.index, 'residual']

```

```
most_extreme_idx = sample_residuals.idxmax() # The index of the highest positive anomaly
(spike)
```

```
# We need the relative position of this index inside the shap_values array
array_position = background_data.index.get_loc(most_extreme_idx)
```

```
print(f"\nGenerating Local SHAP Waterfall Plot for Anomaly at Index
{most_extreme_idx}...")
```

```
print(f"Actual Log Reading: {df_anomalies.loc[most_extreme_idx,
'log_meter_reading']:.2f}")
```

```
print(f"Predicted Log Reading: {df_anomalies.loc[most_extreme_idx,
'predicted_log']:.2f}")
```

```
plt.figure(figsize=(10, 6))
```

```
shap.plots.waterfall(shap_values[array_position], max_display=10, show=False)
```

```
plt.title("Local SHAP Explanation for Extreme Anomaly (Spike)", fontsize=16, pad=20)
```

```
plt.tight_layout()
```

```
plt.show()
```

```
# %%
```

```
import shap
```

```
import pandas as pd
```

```
import numpy as np
```

```
import matplotlib.pyplot as plt
```

```
import seaborn as sns
```

```
from itertools import combinations
```

```
# Ensure SHAP plots render correctly
```

```
plt.style.use('seaborn-v0_8-whitegrid')
```

```
sns.set_context("paper", font_scale=1.2)
```

```
print("--- STARTING MODULE 4: EXPLAINABILITY (RQ2) ---")
```

```
# =====
```

```
# 4.0 LOAD ANOMALIES DATA
```

```
# =====
```

```

file_path = r"C:\Users\rohit\OneDrive - University of East
London\Documents\Assignment\DS7010\Data_Set\energy-
dissertation\data\processed\anomalies_flagged.csv"
df_anomalies = pd.read_csv(file_path, parse_dates=['timestamp'])

# Convert categorical columns back to 'category' datatype for XGBoost compatibility
categorical_cols = ['primary_use', 'meter', 'site_id', 'building_id']
for col in categorical_cols:
    if col in df_anomalies.columns:
        df_anomalies[col] = df_anomalies[col].astype('category')

features = [
    'building_id', 'meter', 'site_id', 'primary_use', 'square_feet',
    'air_temperature', 'cloud_coverage', 'dew_temperature', 'precip_depth_1_hr',
    'sea_level_pressure', 'wind_direction', 'wind_speed',
    'hour', 'day_of_week', 'month'
]

X_test_shap = df_anomalies[features]

# =====
# 4.1 SUBSAMPLE DATA
# =====
print("Subsampling data to prevent memory crash...")
normal_idx = df_anomalies[~df_anomalies['anomaly_iforest']].sample(5000,
random_state=42).index
anomaly_idx = df_anomalies[df_anomalies['anomaly_iforest']].sample(1000,
random_state=42).index

background_data = X_test_shap.loc[normal_idx.union(anomaly_idx)]
print(f"Calculating SHAP values for {len(background_data)} samples... (This may take a
few minutes)")

# =====
# 4.2 INITIALIZE EXPLAINER & PLOT
# =====

```

```

explainer = shap.TreeExplainer(model)
shap_values = explainer(background_data)
print("SHAP calculation complete!")

print("\nGenerating Global SHAP Summary Plot...")
plt.figure(figsize=(12, 8))
shap.plots.beeswarm(shap_values, max_display=10, show=False)
plt.title("Global Feature Importance (SHAP Beeswarm Plot)", fontsize=16, pad=20)
plt.tight_layout()
plt.show()

sample_residuals = df_anomalies.loc[background_data.index, 'residual']
most_extreme_idx = sample_residuals.idxmax()
array_position = background_data.index.get_loc(most_extreme_idx)

print(f"\nGenerating Local SHAP Waterfall Plot for Anomaly at Index
{most_extreme_idx}...")
plt.figure(figsize=(10, 6))
shap.plots.waterfall(shap_values[array_position], max_display=10, show=False)
plt.title("Local SHAP Explanation for Extreme Anomaly (Spike)", fontsize=16, pad=20)
plt.tight_layout()
plt.show()

print("\n--- STARTING MODULE 5: STABILITY METRICS (RQ3) ---")
# =====
# 5.1 EXTRACT TOP-K FEATURES FOR ANOMALIES
# =====
anomaly_mask = df_anomalies.loc[background_data.index, 'anomaly_iforest'] == True
anomaly_indices = background_data[anomaly_mask].index
anomaly_positions = [background_data.index.get_loc(idx) for idx in anomaly_indices]

shap_abs_values = np.abs(shap_values.values[anomaly_positions])
feature_names = background_data.columns

top_k = 3

```



```

anomaly_top_features = []

for i in range(len(anomaly_positions)):
    sorted_feature_idx = np.argsort(shap_abs_values[i])[::-1]
    top_features = set(feature_names[sorted_feature_idx][:top_k])
    anomaly_top_features.append(top_features)

# =====
# 5.2 CALCULATE JACCARD SIMILARITY (CONSISTENCY SCORE)
# =====
def jaccard_similarity(set1, set2):
    intersection = len(set1.intersection(set2))
    union = len(set1.union(set2))
    return intersection / union if union != 0 else 0

jaccard_scores = []
for set1, set2 in combinations(anomaly_top_features, 2):
    score = jaccard_similarity(set1, set2)
    jaccard_scores.append(score)

mean_stability_score = np.mean(jaccard_scores)

print(f"\n--- Explanation Stability Results ---")
print(f"Total Anomalies Analyzed: {len(anomaly_positions)}")
print(f"Mean Top-{top_k} Jaccard Consistency Score: {mean_stability_score:.4f}")

# =====
# 5.3 VISUALIZE STABILITY DISTRIBUTION
# =====
plt.figure(figsize=(8, 5))
sns.histplot(jaccard_scores, bins=20, kde=True, color='forestgreen')
plt.title(f'Distribution of SHAP Explanation Stability (Top-{top_k} Jaccard Score)')
plt.xlabel('Jaccard Similarity Score (1.0 = Perfectly Stable)')
plt.ylabel('Frequency of Anomaly Pairs')
plt.axvline(mean_stability_score, color='red', linestyle='--', label=f'Mean Score: {mean_stability_score:.2f}')

```

```
plt.legend()
plt.tight_layout()
plt.show()

# =====
# 5.4 FEATURE VARIANCE (Which features are unstable?)
# =====
shap_variance = np.std(shap_values.values[anomaly_positions], axis=0)
variance_df = pd.DataFrame({'Feature': feature_names, 'SHAP_Std_Dev': shap_variance})
variance_df = variance_df.sort_values(by='SHAP_Std_Dev', ascending=False)

print("\n--- Feature Explanation Variance (Top 5 Most Volatile) ---")
print(variance_df.head(5).to_string(index=False))
```

2. Data Cleaning Code:

```
# %%
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

# %%
# Set plotting style for publication-ready charts
plt.style.use('seaborn-v0_8-whitegrid')
sns.set_context("paper", font_scale=1.2)

# %%
# =====
# 0. LOAD DATA
# =====
print("Loading data...")
file_path = r"C:\Users\rohit\OneDrive - University of East
London\Documents\Assignment\DS7010\Data_Set\energy-
dissertation\data\processed\ashrae_cleaned_final.csv"
```

```

# %%
# Load data and parse timestamp immediately
df = pd.read_csv(file_path, parse_dates=['timestamp'])
print(f"Dataset loaded successfully! Shape: {df.shape}")

# %%
# =====
# 1.1 BASIC SANITY CHECKS & DISTRIBUTION
# =====
print("\n--- 1.1 Sanity Checks ---")
# Check skewness of the target variable
skewness = df['meter_reading'].skew()
print(f"Skewness of meter_reading: {skewness:.2f}")

# %%
# Check for zero-readings (common anomaly in ASHRAE)
zero_readings = (df['meter_reading'] == 0).sum()
print(f"Percentage of exactly zero readings: {(zero_readings / len(df)) * 100:.2f}%")

# %%
# Plot 1: Distribution of Meter Readings (Log Scale due to high skew in energy data)
plt.figure(figsize=(10, 5))
# Using log1p (log(1+x)) to safely handle zeros
sns.histplot(np.log1p(df['meter_reading']), bins=50, kde=True, color='blue')
plt.title('Distribution of Log-Transformed Meter Readings')
plt.xlabel('Log(1 + Meter Reading)')
plt.ylabel('Frequency')
plt.tight_layout()
plt.show()

# %%
# =====
# 1.2 TIME-SERIES BEHAVIOUR
# =====
print("\n--- 1.2 Time-Series Behaviour ---")

```

```

# Plot 2: Average Energy Consumption by Hour of Day
plt.figure(figsize=(10, 5))
hourly_avg = df.groupby('hour')['meter_reading'].mean().reset_index()
sns.lineplot(data=hourly_avg, x='hour', y='meter_reading', marker='o',
color='darkorange')
plt.title('Average Global Energy Consumption by Hour of Day')
plt.xlabel('Hour of Day (0-23)')
plt.ylabel('Average Meter Reading (kWh)')
plt.xticks(range(0, 24))
plt.tight_layout()
plt.show()

# %%

# Plot 3: Average Energy Consumption by Day of Week
plt.figure(figsize=(10, 5))
daily_avg = df.groupby('day_of_week')['meter_reading'].mean().reset_index()
# Assuming day_of_week is 0-6 (Mon-Sun)
sns.barplot(data=daily_avg, x='day_of_week', y='meter_reading', palette='viridis')
plt.title('Average Global Energy Consumption by Day of Week')
plt.xlabel('Day of Week (0=Mon, 6=Sun)')
plt.ylabel('Average Meter Reading (kWh)')
plt.tight_layout()
plt.show()

# %%

# =====
# 1.3 FEATURE RELATIONSHIPS
# =====

print("\n--- 1.3 Feature Relationships ---")

# Define numerical columns for correlation matrix
numerical_cols = [
    'meter_reading', 'square_feet', 'air_temperature', 'cloud_coverage',
    'dew_temperature', 'precip_depth_1_hr', 'sea_level_pressure',
    'wind_direction', 'wind_speed'
]

```

```
# Calculate Spearman correlation (better for non-linear/skewed data than Pearson)
corr_matrix = df[numerical_cols].corr(method='spearman')

# %%
# Plot 4: Correlation Matrix
plt.figure(figsize=(12, 10))
sns.heatmap(corr_matrix, annot=True, fmt='.2f', cmap='coolwarm', vmin=-1, vmax=1,
            square=True)
plt.title('Spearman Correlation Matrix of Numerical Features')
plt.tight_layout()
plt.show()

# %%
# Plot 5: Weather vs Energy (Air Temperature vs Meter Reading)
# Sampling 10,000 rows for the scatter plot to prevent memory crash/overplotting
sample_df = df.sample(n=10000, random_state=42)

# %%
plt.figure(figsize=(10, 6))
sns.scatterplot(data=sample_df, x='air_temperature', y='meter_reading', alpha=0.3,
               edgecolor=None)
plt.title('Air Temperature vs. Meter Reading (Sampled 10k points)')
plt.xlabel('Air Temperature (°C)')
plt.ylabel('Meter Reading')
plt.tight_layout()
plt.show()

# %%
print(f"Original shape: {df.shape}")

# %%
# Step 1: Remove Exactly Zero Readings (Assuming they are broken sensors/system downtime)
df_pruned = df[df['meter_reading'] > 0].copy()
print(f"Shape after removing zero readings: {df_pruned.shape}")
```

```
# %%  
# Step 2: Remove the extreme outliers seen in your scatter plot  
# (Filtering out readings above 10,000,000 which are likely severe data logging errors)  
df_pruned = df_pruned[df_pruned['meter_reading'] < 10000000]  
print(f"Shape after removing extreme logging errors: {df_pruned.shape}")  
  
# %%  
# Step 3: Apply the Log Transformation to the target variable  
# We do this now so the model trains on the stabilized target  
df_pruned['log_meter_reading'] = np.log1p(df_pruned['meter_reading'])  
  
# %%  
# Step 4: Save the fully prepped data for Module 2  
save_path = r"C:\Users\rohit\OneDrive - University of East  
London\Documents\Assignment\DS7010\Data_Set\energy-  
dissertation\data\processed\ashrae_ml_ready.csv"  
df_pruned.to_csv(save_path, index=False)  
print(f"Fully prepped data saved to: {save_path}")
```

3. Residual Test Analysis Code:

```
# %%  
  
import pandas as pd  
  
import numpy as np  
  
import matplotlib.pyplot as plt  
  
import seaborn as sns  
  
from sklearn.ensemble import IsolationForest  
  
import time  
  
# %%  
  
# Set plotting style  
  
plt.style.use('seaborn-v0_8-whitegrid')
```

```
sns.set_context("paper", font_scale=1.2)

# %%

# =====

# 3.0 LOAD RESIDUALS DATA

# =====

print("Loading baseline test results with residuals...")

file_path = r"C:\Users\rohit\OneDrive - University of East
London\Documents\Assignment\DS7010\Data_Set\energy-
dissertation\data\processed\test_results_with_residuals.csv"

df = pd.read_csv(file_path, parse_dates=['timestamp'])

# %%

# =====

# 3.1 APPROACH A: STATISTICAL THRESHOLD (3-Sigma)

# =====

print("\nCalculating Statistical Anomalies (Mean  $\pm$  3 Std Dev)...")

mean_res = df['residual'].mean()

std_res = df['residual'].std()

upper_bound = mean_res + (3 * std_res)

lower_bound = mean_res - (3 * std_res)

# Flag as True if outside the 3-sigma bounds

df['anomaly_stat'] = (df['residual'] > upper_bound) | (df['residual'] < lower_bound)
```

```
print(f"Statistical Bounds: {lower_bound:.2f} to {upper_bound:.2f}")

print(f"Anomalies detected (Statistical): {df['anomaly_stat'].sum()}
({(df['anomaly_stat'].sum()/len(df))*100:.2f}%")

# %%

# =====

# 3.2 APPROACH B: ISOLATION FOREST

# =====

print("\nTraining Isolation Forest on Residuals... (This may take a minute)")

start_time = time.time()

# We set contamination to 0.01 (assuming ~1% of readings are true anomalies)
iso_forest = IsolationForest(contamination=0.01, random_state=42, n_jobs=-1)

# Fit and predict. It returns -1 for outliers, 1 for inliers.

# We reshape residual to 2D array as required by scikit-learn
predictions = iso_forest.fit_predict(df[['residual']])

# Convert to boolean (True = Anomaly)
df['anomaly_iforest'] = predictions == -1

print(f"Isolation Forest completed in {(time.time() - start_time):.2f} seconds.")

print(f"Anomalies detected (IForest): {df['anomaly_iforest'].sum()}
({(df['anomaly_iforest'].sum()/len(df))*100:.2f}%")

# %%
```



```
# =====  
  
# 3.3 VISUALIZE DETECTIONS ON A SINGLE BUILDING  
  
# =====  
  
print("\nGenerating visual comparison for Building ID: 1258")  
  
# Let's pick a specific building to visualize (1258 is known to have some interesting  
spikes)  
  
# If 1258 isn't in your test set, change this to another ID like 0, 100, or 1000  
  
bldg_df = df[(df['building_id'] == 1258) & (df['meter'] == 0)].copy()  
  
if bldg_df.empty:  
    # Fallback to the first building in the test set if 1258 is missing  
  
    first_bldg = df['building_id'].iloc[0]  
  
    bldg_df = df[(df['building_id'] == first_bldg) & (df['meter'] == 0)].copy()  
  
# Sort by time  
  
bldg_df = bldg_df.sort_values('timestamp')  
  
  
plt.figure(figsize=(15, 6))  
  
# Plot actual energy consumption  
  
plt.plot(bldg_df['timestamp'], np.exp1(bldg_df['log_meter_reading']),  
         label='Actual Energy Usage (kWh)', color='royalblue', alpha=0.6)  
  
# Overlay Statistical Anomalies (Red X)  
  
stat_anomalies = bldg_df[bldg_df['anomaly_stat']]  
  
plt.scatter(stat_anomalies['timestamp'], np.exp1(stat_anomalies['log_meter_reading']),
```

```
color='red', marker='x', s=100, label='Statistical Anomaly', zorder=5)

# Overlay Isolation Forest Anomalies (Orange Circles)
iforest_anomalies = bldg_df[bldg_df['anomaly_iforest']]
plt.scatter(iforest_anomalies['timestamp'],
            np.exp1(iforest_anomalies['log_meter_reading']),
            color='orange', marker='o', s=80, facecolors='none', linewidth=2,
            label='Isolation Forest Anomaly', zorder=4)

plt.title(f"Anomaly Detection Comparison: Building {bldg_df['building_id'].iloc[0]}
(Electricity)")
plt.xlabel("Date")
plt.ylabel("Energy Consumption (kWh)")
plt.legend()
plt.tight_layout()
plt.show()

# %%

# Save the dataset with anomaly flags for Module 4 (SHAP Explainability)
save_path = r"C:\Users\rohit\OneDrive - University of East
London\Documents\Assignment\DS7010\Data_Set\energy-
dissertation\data\processed\anomalies_flagged.csv"
df.to_csv(save_path, index=False)
print(f"Saved anomaly flags to: {save_path}")
```